

Chapter 4

Ultra-High-Speed GaAs VLSI: Approaches, Potential, and Progress*

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I. INTRODUCTION

The achievement of ultra-high-speed VLSI, that is, of integrated circuit chips with complexities of $N_g > 10^4$ equivalent logic gates with propagation delays of $\tau_d \approx 100$ psec or less, would make possible computational powers or chip throughput rates hard to imagine by today's standards. The realization of such potential is very difficult, of course, since it necessitates achieving in one circuit and fabrication technology: (1) ultra high speed (very low τ_d), (2) low power per gate (P_D), (3) extremely low dynamic switching energies (power-delay products $P_D\tau_d$), (4) very high gate densities, and (5) very high process yields (sufficient to allow economic fabrication of such complex parts) [1,2]. The problem here is the requirement for improvements on both sides of what are classical trade-offs such as speed-power or lithographic resolution-yield. For example, in silicon MOS, the speed may improve by increasing the supply voltage and logic voltage swings in order to increase the average device transconductances or current gain-bandwidth products (f_T). Increasing V_{DD} , however, while reducing τ_d , sharply increases gate power dissipation P_D and switching energies $P_D\tau_d$, which would lead to unacceptable power levels in $>10^4$ gate VLSI chips [1,2]. Reducing geometry by pressing lithographic resolution is an approach to coping with this speed-power trade-off, and it improves density as well, but this approach can easily result in unacceptable reduction in yield if pressed too far.

Some of the aspects of these trade-offs are illustrated in Table I, which gives a review of the circuit requirements for such ultra-high-speed, low-power logic, and their consequences in terms of device requirements if such circuits are to be realized. Clearly, since logic gates classically must have current and power gains greater than their fan-out loadings, the f_T and f_{max} (current and power gain-bandwidth products) of the active devices must be very high. Perhaps slightly less obviously, the devices must have a high degree of nonlinearity, i.e., they must develop their high transconductance (or f_T or f_{max}) at control voltages only a small logic volt-

TABLE I

**Circuit and Consequent Device Requirements for Very-High-Speed,
Low-Power Logic Circuits**

CIRCUIT REQUIREMENT	CONSEQUENT DEVICE REQUIREMENTS
1. LOW LOGIC VOLTAGE SWINGS (FOR LOW POWER $\sim \frac{1}{2}C\Delta V^2$)	VERY UNIFORM THRESHOLD VOLTAGES FOR ACTIVE DEVICES (PARTICULARLY FOR VLSI)
2. LOW CAPACITANCES (BOTH DEVICE AND PARASITIC)	LOW INPUT CAPACITANCE DEVICES HIGH CIRCUIT DENSITIES SEMI-INSULATING SUBSTRATE FOR LOW CIRCUIT PARASITICS
3. HIGH SWITCHING SPEEDS WITH REASONABLE FANOUT LOADINGS AT LOW SWITCHING VOLTAGES	VERY HIGH f_T (CURRENT GAIN BANDWIDTH) VERY HIGH f_{max} (POWER GAIN BANDWIDTH) VERY RAPID INCREASE IN TRANSCONDUCTANCE ABOVE THRESHOLD (STRONG NONLINEARITY)
4. HIGH CIRCUIT COMPLEXITIES WITH ACCEPTABLE COST (HIGH YIELDS)	EASILY FABRICABLE DEVICE STRUCTURES HIGH YIELD PROCESS THRESHOLD VOLTAGES AND OTHER CRITICAL DEVICE PARAMETERS INSENSITIVE TO REASONABLE PROCESSING VARIATIONS TIGHT, PREDICTABLE PARAMETER STATISTICS
5. DELIVERY ASAP	ESTABLISHED FABRICATION TECHNOLOGY

age swing above threshold. This is necessitated by the need to drastically reduce dynamic switching energies ($P_{D\tau_d}$) for ultra-high-speed VLSI. The dynamic switching energy must exceed the stored energy on the switched capacitance C as

$$P_{D\tau_d} > \frac{1}{2}C(\Delta V_L)^2, \quad (1)$$

where C is the sum of the input capacitances of the fan out of loading gates plus the parasitic capacitance and ΔV_L the logic voltage swing. (More precisely [1,2], the $(\Delta V_L)^2$ term is the product of the supply voltage V_{DD} times ΔV_L). Hence, the power supply and logic swing voltages must be kept small for low power-delay products, and the devices must not only be fast but must be fast with small logic voltage swings. A further consequence of this requirement for small logic voltage swings is that the active devices have threshold voltages that are very precisely controlled, i.e., with standard deviation of threshold voltage a small fraction (preferably $< 5\%$ or better) of the logic voltage swing.

II. GaAs DEVICE APPROACHES FOR ULTRA-HIGH-SPEED VLSI

As illustrated in Table I, the circuit constraints for ultra-high-speed, low-power logic for VLSI generate a number of requirements for the

TABLE II

**Device Characteristics Desired for Optimal High-Speed Low-Power Switching with the
Consequent Device Physical Parameters and Structural Characteristics**

DESIRED ELECTRICAL CHARACTERISTIC	CONSEQUENT PHYSICAL PARAMETERS
HIGH TRANSCONDUCTANCE AT CONTROL VOLTAGES A SMALL SWING ABOVE THRESHOLD	HIGH CARRIER MOBILITIES EXTREMELY SHORT CONTROL TRANSIT PATH (e.g., GATE LENGTH OR BASE WIDTH)
VERY UNIFORM THRESHOLD VOLTAGES, V_T (TO ALLOW LOW VOLTAGE SWING LOGIC)	EXTREMELY LOW V_T SENSITIVITY TO <u>HORIZONTAL</u> GEOMETRY VARIATIONS LOW V_T SENSITIVITY TO REASONABLE <u>VERTICAL</u> GEOMETRY VARIATIONS LOW V_T SENSITIVITY TO <u>DOPING</u> VARIATIONS
VERY LOW INPUT CAPACITANCES	SMALL GEOMETRIES (e.g., GATE LENGTH AND WIDTH IN FETs OR EMITTER AREA IN BIPOLARS)
HIGH f_T AND f_{max} GAIN-BANDWIDTH PRODUCTS	HIGH CARRIER MOBILITIES AND SATURATION VELOCITIES, SMALL GEOMETRIES (SHORT TRANSIT LENGTHS; e.g., GATE LENGTH, BASE WIDTH, OR EMITTER STRIPE WIDTH)
HIGH YIELD	SIMPLE, EASILY FABRICATED STRUCTURE

characteristics of the active devices needed to implement such circuits. These required device characteristics also have numerous implications for the general physical characteristics needed for the devices themselves. These considerations can be very useful in selecting promising device approaches for ultra-high-speed VLSI. In Table II some of the desired device characteristics from Table I are reviewed with focus on their consequences in terms of the physical parameters required for the devices (e.g., structures, geometries, and semiconductor parameters).

A. Importance of Mobility for VLSI

As indicated in Table II, a key consequence of the requirement for strong nonlinearity, that is, high transconductance g_m , and f_T at control voltages only small logic swings above threshold is the need for high carrier mobilities and/or very short carrier transit path lengths in the devices. This is easily illustrated for an FET structure [1,2], where for gate voltages V_{gs} near threshold V_p , the saturated drain current I_{ds} is given by

$$I_{ds} = K(V_{gs} - V_p)^2 = (\epsilon\mu_n W/2aL_g)(V_{gs} - V_p)^2, \quad (2)$$

where W is the channel width, L_g the gate length (the distance between gate and channel), μ_n the electron mobility (assuming n -channel), ϵ the dielectric constant of the semiconductor in a MESFET or JFET or of the

gate insulator in an IGFET (MOSFET), and a the spacing between the gate and the channel. The transconductance g_m for this Shockley FET model is then

$$g_m = (\epsilon\mu_n W/aL_g)(V_{gs} - V_p), \quad (3)$$

and the current gain–bandwidth product in this near-threshold, small $V_{gs} - V_p$ region is given by

$$f_\tau = g_m/2\pi C_{gs} = (\mu_n/2\pi L_g^2)(V_{gs} - V_p). \quad (4)$$

Equation (3) illustrates that the strength of the nonlinearity, that is, the increase of transconductance with small gate voltages above threshold increases as μ_n/L_g , while the f_τ expression of Eq. (4), which ignores fringing capacitances, gains an additional $1/L_g$ dependence due to the assume capacitance variation. (In real short-channel devices, the capacitance variation is much slower than $1/L_g$ because of the fringing effects.)

Clearly, to maximize the transconductance at any given $V_{gs} - V_p$ voltage swing one can reduce the gate length L_g to shorter and shorter values. Unfortunately, taking this course alone places unreasonable pressures on the required lithographic precision and will ultimately prove disastrous to yield. Note, however, that we can obtain an equivalent improvement in near-threshold transconductance by going to a semiconductor with a higher channel mobility μ_n . This is, of course, the prime motivation for using GaAs rather than silicon in high-speed FETs and integrated circuits. Electron mobilities in GaAs of several hundred thousand have been obtained in undoped material at 77 K (8500 cm²/V sec at room temperature), while in typical MESFET n -channel dopings ($N_d \approx 10^{17}$ cm⁻³) electron mobilities in the $\mu_n \approx 4000$ –5000 cm²/V sec range are obtained (see Fig. 1). These GaAs electron mobilities are about six times higher than those for correspondingly doped bulk silicon and nearly an order of magnitude larger than typical silicon MOS n -channel mobilities under strong inversion conditions. As a consequence, very high f_τ s and switching speeds can be obtained in GaAs short-channel devices even with small voltage swings ΔV_L . For example, at room temperature, $L_g = 0.6$ μ m, e-beam-fabricated, GaAs E-MESFET ring oscillators have given $\tau_d = 30$ psec delays at $\Delta V_L \approx 0.5$ V, while at 77 K (where the electron mobility is even higher), $\tau_d = 17$ psec speeds were measured in the F1 = F0 = 1 ring oscillators¹ [3].

¹ Data is presented for $L_g = 0.6$ μ m, 15-stage, inverter ring oscillators with room temperature speeds up to $\tau_d = 30$ psec at $P_D = 1.9$ mW/gate or $\tau_d = 63$ psec at $P_D = 100$ μ W/gate, and 77 K speeds up to $\tau_d = 17.5$ psec at $P_D = 9.2$ mW/gate. Earlier $L_g = 0.8$ μ m data is presented in [3a].

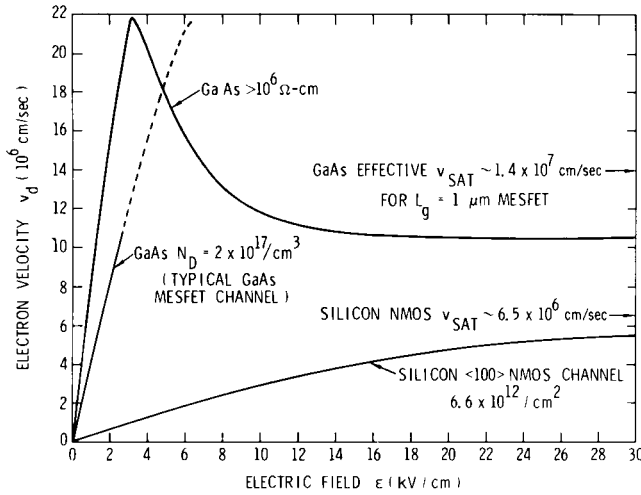


Fig. 1. Comparison of the equilibrium velocity-field characteristics for electrons in silicon MOS channels and GaAs at 300 K. The high-resistivity GaAs exhibits larger mobilities than typical GaAs MESFET channels (the curve for which is simply extrapolated from the low field mobility).

B. Supermobility Devices: The HEMT

While these 77 K E-MESFET results are impressive, the electron mobilities in the channel at 77 K are not too much higher than at room temperature due to electron Rutherford scattering off of ionized donor impurities (typically $N_D^+ \approx 10^{17}$ in MESFET channels). Special GaAs/GaAl_xAs_{1-x} FET structures variously referred to as modulation-doped FETs or high electron mobility transistors (HEMTs) [4] should greatly improve this 77 K channel mobility. The key in these structures is to place the donor atoms in a wider-bandgap GaAl_xAs_{1-x} layer adjacent to an undoped GaAs channel layer, which in fact receives the free electrons from the ionized donors. Electron channel mobilities of $\mu_n = 49,300$ cm²/sec at 77 K have been reported for HEMT structures [5]. From Eqs. (3) and (4), such FETs, fabricated with reasonably short gate lengths, should achieve high g_m and F_T with logic swings of only $\Delta V \approx 100$ mV or so.

C. Saturation and Ballistic Transport Effects

It is important to note the limitations of Eqs. (2)–(4), particularly when applied to very short channel devices operating at substantial voltage swings. If we naively used Eq. (4) to estimate the f_T for an $L_g = 0.5$ μm

HEMT with $\mu_n = 5 \times 10^4 \text{ cm}^2/\text{V sec}$ at $V_{gs} - V_p = 5 \text{ V}$, we would calculate an $f_T = 1.6 \times 10^{13} \text{ Hz}$ (16,000 GHz!). However, by these same assumptions, the electron velocity in the channel would be

$$v_d = \mu_n \epsilon_c \approx \mu_n (V_{gs} - V_p) / L_g, \quad (5)$$

or $v_d = 5 \times 10^9 \text{ cm/sec}$, for the conditions assumed. Inasmuch as the velocity of light in GaAs is only $9 \times 10^9 \text{ cm/sec}$, this is, to say the least, suspiciously high (as is, of course, the calculated f_T). The problem with our naive calculation was, of course, that we have assumed ohmic, $v_d = \mu_n \epsilon_c$, behavior at any ϵ_c , while in fact the equilibrium velocity-field curves, as shown in Fig. 1, show velocity saturation behavior at high field strengths. One must hasten to point out, however, that it would be just as naive to assume that in very short channel, low-voltage MESFETs or HEMTs that the equilibrium GaAs velocity-field characteristics of Fig. 1 would rigorously apply. If we took, for example, the previous case of an $L_g = 0.5 \mu\text{m}$ HEMT with $\mu_n = 5 \times 10^4 \text{ cm}^2/\text{V sec}$, only now operating at a more reasonable $V_{gs} - V_p = 250 \text{ mV}$, we would have $\epsilon_c = (V_{gs} - V_p) / L_g = 5000 \text{ V/cm}$. This would give, from Eq. (5), $v_d = 2.5 \times 10^8 \text{ cm/sec}$, as based on the ohmic assumption, which is clearly too high, since a purely ballistic (kinetic energy = electrostatic energy) electron of mass m_e dropping through a potential of $V_{gs} - V_p = 250 \text{ mV}$ would have a velocity of only

$$v_B = [2(\text{kinetic energy})/m_e]^{1/2} = [2q(V_{gs} - V_p)/m_e]^{1/2}, \quad (6)$$

or $v_B = 1.1 \times 10^8 \text{ cm/sec}$, for $m_e = 0.072$ times the free electron mass (actually v_B would be slightly less than this as the GaAs electron effective mass increases somewhat away from the conduction band minimum).

While the finite electron mass puts an upper limit of about $v_d = 10^8 \text{ cm/sec}$ on the electron velocity in our $V_{gs} - V_p = 250 \text{ mV}$, $L_g = 0.5 \mu\text{m}$ HEMT example, the equilibrium static velocity-field curve for GaAs at $\epsilon_c = 5000 \text{ V/cm}$ gives a velocity of only $v_d = 1.8 \times 10^7 \text{ cm/sec}$. (This would drop even lower at higher field strengths, approaching $v_{sat} \approx 1 \times 10^7 \text{ cm/sec}$.) That this equilibrium velocity-field projection is completely unreasonable for this example is clear when one considers *why* the GaAs velocity-field curve shows the peak and negative differential mobility behavior (finally leading to the $v_{sat} \approx 1 \times 10^7 \text{ cm/sec}$ saturation) that it does. The abrupt reduction in mobility and electron velocity in GaAs at $\epsilon_c \approx 3200 \text{ V/cm}$ is caused by the transfer of electrons from the very narrow low m_e conduction band minimum at Γ ($K = \langle 000 \rangle$) in GaAs to a plurality of high mass, siliconlike electron minima lying about $\Delta E = 0.3 \text{ eV}$ above the Γ minimum. Electrons in the higher valleys have high mass and strong intervalley scattering and thus exhibit very low mobility (like

conduction electrons in silicon). However, note that because these minima lie at $\Delta E = 0.3$ eV above the Γ minimum and we have assumed only a $V_{gs} - V_p = 0.25$ V maximum potential across the channel for this example, it is physically impossible for electrons to transfer to these low-mobility valleys. Hence, the equilibrium $v_d - \epsilon_c$ curve of Fig. 1, measured for long devices with tens of volts across them, is completely irrelevant for this low-voltage, short-channel HEMT case. In fact, even for $V_{gs} - V_p > 0.3$ V operation in very short ($L_g \approx 0.5 \mu\text{m}$) channel GaAs FETs, the electron transit times through the channel are so short (approaching 1 psec) that electron transfer to higher valleys is not completed during transit. These "velocity overshoot," or "ballistic," effects are predicted to improve the performance of all short transit path GaAs devices, particularly when the operating voltages are low.

D. Effect of Velocity Saturation on FET Performance

Based on the preceding arguments, we expect that in our $L_g = 0.5 \mu\text{m}$ short-channel 77 K HEMT example, electron velocities will be well above the $v_d = 1.8 \times 10^7$ cm/sec equilibrium $v_d - \epsilon_c$ curve (Fig. 1) prediction but definitely below the $v_B = 1.1 \times 10^8$ cm/sec ballistic limit for $V_{gs} - V_p = 0.25$ V (and certainly far below the $v_d = 2.5 \times 10^8$ cm/sec predicted from $v_d = \mu_n \epsilon_c$, ignoring the effect of electron mass). Reasonable estimates from calculations and diode measurements [6] of the transient electron velocity in GaAs under conditions similar to this example are around $v_d \approx 5 \times 10^7$ cm/sec. This is as compared to saturation velocities of $v_{sat} \approx 6.5 \times 10^6$ cm/sec in bulk silicon (at very high ϵ_c fields, approaching 10^5 V/cm). If we assume saturated electron velocity ($v_d = v_{sat}$, rather than $v_d = \mu_n \epsilon_c$), Eqs. (3) and (4) become

$$g_m = \epsilon v_{sat} W/a \quad (\text{for } v_d = v_{sat}) \quad (7)$$

for transconductance in this velocity-saturation region and

$$f_r = v_{sat}/2\pi L_g \quad (\text{for } v_d = v_{sat}) \quad (8)$$

for the current gain-bandwidth product. By this calculation, the $0.5\text{-}\mu\text{m}$ HEMT with $a = 0.1 \mu\text{m}$ at $V_{gs} - V_p < 0.3$ V could have g_m/W values approaching 400 mS/mm and f_r values from Eq. (8) near 100 GHz for effective v_{sat} values approaching 4×10^7 cm/sec (as compared to rather unrealistic $f_r = 800$ GHz and $g_m/W = 5300$ mS/mm values from Eqs. (4) and (3) assuming $v_d = \mu_n \epsilon_c$). More realistic capacitance estimates, including fringing, would probably roughly halve the 100-GHz f_r estimate for the ($\mu_n = 50,000$) HEMT to perhaps 50 GHz. This is not a great deal higher (perhaps two times) than would be expected for a normal ($\mu_n = 4000$)

$L_g = 0.5 \mu\text{m}$, 300 K GaAs MESFET, but the normal FET would require five to ten times higher voltage swings to achieve full f_T . As a consequence, the HEMT could, in principle, achieve somewhat higher switching speeds in 77 K operation than could a normal GaAs MESFET of similar geometry but at far lower logic swings so that the dynamic switching energies might be one or two orders of magnitude less. This would, of course, be very attractive for VLSI.

E. Limitations to Very Low Logic Voltage Swing Operation

Achieving these exceedingly low $P_{D\tau_d}$ products in very short channel HEMTs (or other devices) by operating with very low logic swings ($\Delta V \approx 100 \text{ mV}$ or less) assumes that such an operation is in fact practical from a circuit standpoint. Given that the device nonlinearities are strong, as is the case for the HEMT, the limitations on logic swing are that ΔV_0 must be large in comparison to circuit noise levels and large in comparison to key parameter variations, such as threshold voltage. The thermal noise level $V_n = kT/q$ is only 6.64 mV at 77 K, and logic swings of $10 kT/q$ or higher are adequate, so that $<100\text{-mV}$ swings are thermodynamically acceptable at 77 K. A more serious “noise” problem is the “hash,” or transient noise spikes, from high-speed switching operations electromagnetically coupled onto the circuit power supply and ground conductors as well as directly to other digital signal lines. While, in principle, this “hash” can be made arbitrarily small by careful electromagnetic design of the integrated circuit layouts, packages, and interconnects, in fact, as switching speeds increase, the $L di/dt$ or $M di/dt$ and $C dv/dt$ terms all tend to make this problem worse. It is not unreasonable to expect that switching noise or “hash” levels could become a limiting factor in how low the logic swings may be reduced in practical ultra-high-speed VLSI circuits.

A much more severe limitation to the reduction of logic voltage swing in GaAs MESFET or HEMT circuits is the minimum variation of threshold voltage V_p of the FETs over a circuit of wafer achievable in the fabrication process. As will be discussed later, in order to accommodate the statistical variations in V_p without unacceptable degradation of dynamic switching characteristics, the logic voltage swing ΔV_L must be at least ~ 20 times the standard deviation of V_p (preferably higher). This would mean that for the HEMT example of $\Delta V_L = 100 \text{ mV}$, a $\sigma_{V_p} < 5 \text{ mV}$ uniformity would be required for the FETs. As will be discussed later, the best FET V_p uniformities achieved over a GaAs wafer are $\sigma_{V_p} = 34 \text{ mV}$, with $\sigma_{V_p} = 50 \text{ mV}$ more reasonable [7] (see Fig. 10), much too high for 100 mV super-low logic swing operation. Hence, attaining ultra-low

dynamic switching energies through very low logic swing circuit operation will necessitate either major improvements in materials and processing techniques to allow exceedingly tight control of FET pinch-off voltage or else the use of a device approach in which the threshold voltage is inherently extremely uniform.

F. The GaAs Heterojunction Bipolar Transistor for Superspeed VLSI

The realization of very low logic voltage swings requires a high degree of nonlinearity (rapid increase in transconductance for small voltage swings ΔV_L above the threshold) combined with extremely uniform threshold voltage devices, which probably means device structures in which the threshold voltage is highly insensitive to normal processing parameter variations (horizontal dimensions, vertical dimensions, doping level variations, etc.). While the GaAs MESFET pinch-off voltage is insensitive to horizontal geometry variations (e.g., L_g), V_p is sensitive to both the thickness (vertical geometry) and doping level in the channel layer. In comparison, an almost ideal device from the standpoint of threshold voltage variations is the bipolar transistor. In a bipolar transistor, V_{BE} varies linearly with temperature T (which can presumably be kept quite uniform over a chip) but only logarithmically with base width W_B , base doping level N_A , mobility μ_n , collector current I_c , etc. Threshold uniformities $\sigma_{V_{BE}}$ of a few millivolts over a circuit or wafer are fairly easily achieved in silicon bipolar transistors.

In spite of the inherent threshold uniformity of bipolar and the great success of silicon (homojunction) bipolar transistors in high-speed logic (such as ECL), the homojunction bipolar transistor involves some very unfortunate inherent design trade-offs, which drastically limit performance and make it quite unattractive for GaAs implementation. These design compromises center on the fact that to maintain emitter injection efficiency, the base doping level N_A must be limited to a fraction of a percent of the emitter doping. Furthermore, the active (transiting) carrier is the minority carrier in the base (e.g., electrons in an NPN bipolar), and it is up to the rather modest density N_A of holes in the base to establish electrical contact from the control electrode to the active region with acceptably small lateral base resistance R_B . The inherent performance of an NPN bipolar improves strongly as the electron transit time through the base is reduced by lowering the base width W_B , but this leads immediately to unacceptably high levels of base resistance R_B unless the emitter stripe width W_E is exceedingly small (e.g., $\ll 1 \mu\text{m}$). Further, if we lower W_B too far (e.g., $< 1000 \text{ \AA}$), with the modest base doping levels involved in

homojunction bipolars, the numbers of doping atoms in a cube of dimension W_B on each side get so small ($\sim 10^2$) that simply the Gaussian statistical variations in this number can lead to emitter–collector punchthrough in a statistically significant (for VLSI) number of transistors. As far as GaAs is concerned, the fact that the homojunction bipolar transistor requires both high electron mobility μ_n for short transit time *and* high hole mobility μ_p for low R_B is a problem because the hole mobility in GaAs is even less than that in silicon. Hence there is little reason to expect that the performance of a homojunction bipolar transistor in GaAs would be much better than in silicon.

While homojunction bipolar transistors in GaAs offer few apparent advantages over silicon, the availability of high-quality $\text{Ga}_{1-x}\text{Al}_x\text{As}$ to GaAs heterojunctions, as used in the HEMT FETs, for example, makes it possible to fabricate heterojunction bipolar transistors (HJBT) that avoid most of the disadvantageous trade-offs of homojunction bipolars [8, 9]. The heterojunction bipolar transistor makes use of a wider band-gap emitter (typically $10kT/q$ larger than that of the base, or ~ 0.25 eV for room temperature use), such as n -type $\text{Ga}_{1-x}\text{Al}_x\text{As}$ emitter on a heavily doped p^{2+} GaAs base in an NPN HJBT. The larger emitter band gap makes hole injection from the base into the emitter virtually impossible, even when the base doping level is much higher than that in the emitter. Hence, in the HJBT, there is no trade-off between maintaining β or emitter injection efficiency and increasing base doping N_A , as required to reduce base width W_B without increasing the lateral base resistance R_B . In addition, the high N_A minimizes base depletion (Early) and punchthrough effects from the collector bias and the higher emitter doping level reduces the emitter–base capacitance. Like the homojunction bipolar transistor, the current flow is vertical, hence the f_T -limiting transit path distance (the base width W_B) is limited only by our ability to make thin semiconductor layers, and molecular-beam epitaxial (MBE) layers tens of angstroms in thickness have been demonstrated. Unlike homojunction bipolars, though, the heavy base dopings in HJBTs allow practical use of base thicknesses of a few hundred angstroms.

In nonsaturating (ECL-type) logic circuits, the HJBT may be used with a GaAs homojunction collector. However, if this structure is allowed to go into saturation (excessive I_B with strongly forward-biased collector–base junction), hole injection from the base into the collector region will result in a large minority carrier stored charge in the collector, which will slow turn-off unacceptably, just as in silicon bipolar transistors operated in saturated switching. Fortunately, in the HJBT, it should be possible to use a wide band-gap collector region (along with emitter) to confine the excess stored charge in saturated operation to the very thin base region only.

This should make it possible to use, for very high speed VLSI, such simple, very low power, high-density saturated logic approaches as I²L. Of course, higher speeds would be expected for nonsaturated logic approaches (ECL, CSL, etc.) but at somewhat higher power levels and lower gate densities.

The heterojunction bipolar transistor (HJBT), from both an ultimate performance and a parameter statistics standpoint, appears to be ideal for ultra-high-speed VLSI. Even with comparatively modest (1- μ m) emitter stripe widths and comfortable (for MBE) 500-Å base widths, f_T values of the order of 100 to 200 GHz (or intrinsic transistor switching speeds of ~ 1 psec) should be attainable at reasonable emitter current densities, and this is at room temperature. Because, as with the homojunction bipolar transistor, the "threshold voltage" (V_{BE} for some specific I_C) is determined mainly by the energy gaps of the base and emitter regions, rather than geometrical and doping factors, the threshold uniformity should be outstanding ($\sigma_{V_{BE}}$ approximately equal to a few millivolts). Consequently, the logic swings ΔV may be reducible to the basic thermal noise or switching "hash" limits for ultra low dynamic switching energy operation. In the ECL-type circuit approaches, the balanced current operation tends, with proper layout and design rules, to minimize the switching noise, or "hash," so that very low ($\Delta V \approx 10kT/q$) logic swing operation may prove practical in HJBT ECL circuits).

If f_T s of 200 GHz indeed prove achievable in 1- μ m emitter geometry heterojunction bipolar transistors with low parasitics (i.e., semi-insulating GaAs substrate), then an $F_0 = F_1 = 1$ HJBT ring oscillator could show logic delays of under 2 psec or, more reasonably loaded circuit propagation delays in the $\tau_d = 5\text{--}10$ psec range. At these propagation delays, on-chip interconnect delays (~ 10 psec/mm) could be quite significant, and interconnect lines longer than a few millimeters would probably have to be driven and terminated in their characteristic impedance to avoid reflections. Of course, the high transconductances in the HJBTs should readily facilitate achieving typical line impedances in emitter-follower drivers or active terminations in minimally sized devices.

In summary, this type of GaAs heterojunction bipolar transistor circuit technology could achieve, at room temperature of 77 K, logic speeds as high or higher than those of Josephson junction circuits at 4 K. Unfortunately, very little work relevant to ultra-high-speed VLSI has been done on HJBTs. While some work on GaAs/Ga_{1-x}Al_xAs heterojunction bipolars for microwave amplifier applications has been published [10], most of the very thin base, high f_T MBE or MOCVD-fabricated HJBT work has been oriented toward phototransistor applications (see references cited in Bailbe *et al.* [10]). A complete fabrication technology will have to be de-

vised before this highly attractive device approach can be brought to bear on the achievement of ultra-high-performance logic for VLSI.

G. Comparison of Prospective GaAs Device Approaches for VLSI

In Table I some of the circuit requirements for ultra-high-speed, low-power logic for VLSI were outlined and interpreted in terms of desirable material qualities, such as semi-insulating substrates for low parasitics, and device requirements, such a very high f_T at low voltage swings and very uniform threshold voltages. In Table II these device requirements for ultra-high-speed VLSI were reviewed in terms of the device structural features and semiconductor parameters (e.g., electron mobilities and velocities) necessary to realize them. The previous five sections have illustrated these device requirements by focusing on specific examples in which their roles are clear. In Table III this discussion is summarized and extended to a comparison of most of the major contenders in GaAs device technologies for ultra-high-speed, low-power logic for VLSI from the

TABLE III

Comparison of GaAs Device Approaches for High-Speed Low-Power Logic Circuits

DEVICE TYPE	NONLINEARITY $HI_{gm} @ L_0 V_G - V_T$	THRESHOLD VOLTAGE UNIFORMITY				f_T	FABRICABILITY, YIELD
		COMBINED	HORIZONTAL SENSITIVITY	VERTICAL SENSITIVITY	DOPING SENSITIVITY		
HETERO-JUNCTION BIPOLAR TRANSISTOR ($W_E \sim 1 \mu m$, $W_B < 0.1 \mu m$)	EXCELLENT (KT LIMITED) ($gm @ V_{BE} - V_T$ $\sim 100 mV$)	EXCELLENT (ENERGY GAP) $\sigma V_T \sim 1 mV$	EXC.	EXC.	EXC.	EXCELLENT (100-200 GHz)	UNKNOWN - NO PROCESS DEVELOPED
GaAs MESFET ($L_g \sim 1 \mu m$, $a \sim 0.1 \mu m$)	GOOD ($gm @ V_{gs} - V_p$ $\sim 1V$)	OK ^a (DOPING PROFILE) $\sigma V_T \sim 40 mV$	EXC.	FAIR (ADEQUATE WITH I.I.)	FAIR	GOOD ($\sim 15 GHz$)	EXCELLENT, SIMPLE, HIGH YIELD
GaAs MESFET ($L_g \sim 0.5 \mu m$, $a \sim 0.1 \mu m$)	GOOD - VERY GOOD ($gm @ V_{gs} - V_p$ $\sim 0.5V$)	OK ^a (DOPING PROFILE)	EXC.	FAIR (ADEQUATE WITH I.I.)	FAIR	VERY GOOD ($\sim 30 GHz$)	DIFFICULT LITHOGRAPHY
HETERO-JUNCTION FET (HIGH MOBILITY) ($L_g = 1 \mu m$, $a \sim 0.1 \mu m$, $T = 77^\circ K$)	EXCELLENT WHEN COOLED ($gm @ V_{gs} - V_p$ $\sim 100 mV$)	OK? (DOPING PROFILE)	EXC.	FAIR (ADEQUATE WITH MBE?)	FAIR	VERY GOOD ($\sim 20 GHz$ OR $50 @ 0.5 \mu m$?)	MBE FABRICATION, NEW PROCESS
"PERMEABLE BASE" TRANSISTOR, PRESENT METAL FINGER SIZE ($0.16 \mu m$)	VERY GOOD ($gm @ V_{gs} - V_p$ $\sim 0.3V$)	VERY POOR (HIGHLY GEOMETRY SENSITIVE)	POOR (FINGER SPACING)	FAIR	POOR	GOOD ($17 GHz f_{max}$ MEASURED WITH $37 GHz f_T$)	DIFFICULT EPI PROCESS AND FINGER LITHOGRAPHY
FUTURE "PERMEABLE BASE" TRANSISTOR @ 500Å FINGER SIZE	VERY GOOD	VERY POOR	POOR	FAIR	POOR	EXCELLENT (200 GHz f_T CALCULATED)	NO KNOWN PROCESS FOR THIS FINGER SIZE

^a MESFET threshold uniformities satisfactory for DMESFET logic with $\sim 1-V$ logic swings, very marginal for 0.5-V swing EMESFET logic.

“plain vanilla” (e.g., $L_g = 1\text{-}\mu\text{m}$ MESFET) to the “far out” (e.g., 5–10-psec switching heterojunction bipolar transistors). Reviewed in Table III are performance factors such as the ultimate speed capabilities (as limited by f_T or $f_{\max}$) and the ultimate capability to achieve those high speeds with very low dynamic switching energies (low logic voltage swings as determined by the nonlinearity or voltage swing necessary to achieve high g_m or f_T).

On a more practical basis, however, the attainment of low voltage swing operation requires a very tight threshold voltage distribution for the switching devices. Hence, in Table III, the device approaches are rated according to their inherent sensitivities to horizontal device dimension variations (which are the worst since these are caused directly by the lithographic process and are most difficult to control), to vertical device dimension variations (e.g., implant depth, epitaxial layer thicknesses), and to doping density variations in the active regions of the devices.

Also shown in Table III is a “combined” overall threshold voltage uniformity rating estimated for each device type. These should be considered as tentative, inasmuch as only the horizontal (lithographic) variations are fundamental and common to all device types, and most of devices are insensitive to these horizontal variations. The vertical geometry and doping density variations are highly fabrication-approach and fabrication-technology dependent. For example, GaAs DMESFETs fabricated by liquid-phase epitaxy (LPE) in the early to mid-1970s had standard deviations of V_p of the order of 1 V, principally due to layer thickness variations. In our planar ion-implanted GaAs IC process [1,2,7], we have obtained DMESFETs with standard deviations over a wafer as low as 34 mV (Fig. 10) by careful control of implant range and dose and the compensating impurity level in the substrate material in the region of the active device channels. Molecular-beam epitaxy is capable of fabricating very uniform thickness layers and hence should also be capable of uniform V_p s. Hence, either changes of the method by which these devices are fabricated or improvements in the particular fabrication technologies involved can produce improvements in the threshold voltage uniformities.

These threshold voltage uniformity projections can, in spite of their technology-dependent nature, be useful in weighing prospective device approaches for ultra-high-speed VLSI. For example, the heterojunction bipolar transistor shows outstanding f_T and nonlinearity, in combination with inherent outstanding threshold voltage uniformity. The permeable-base transistor [11], a vertical-geometry structure like an NPN bipolar transistor but with the p -type base replaced with a very fine metal grid (similar to Shockley’s analog triode structure), also should, with small enough grid finger geometries ($\sim 500\text{-}\text{\AA}$ lines and spaces), be able to achieve 200-GHz cutoff frequencies [11] and have very good nonlinearity.

Unfortunately, in this device structure, the pinch-off voltage is not only dependent on the vertical dimensions (epitaxial layer thicknesses) and donor concentrations, but it also varies strongly with a horizontal (lithographically generated) dimension, the spacing between fingers. Inasmuch as the finger lithography (1600-Å lines and spaces, fabricated by laser interference in current $f_r = 37$ GHz, $f_{\max} = 17$ GHz experimental devices) is at the edge of the state of the art, it seems unreasonable to expect that tight control of threshold voltages in these devices will be realized. Hence, their use in low-power (low logic voltage swing), ultra-high-speed circuits, particularly at the VLSI complexity level, appears highly unlikely. These comments also apply to other, similarly gridded structures dimensioned to take advantage of ballistic transport (transient velocity overshoot) effects in GaAs.

The last column in Table III, while perhaps mundane from a device physics point of view, is probably the most important of all for VLSI: fabricability and yield potential. The enormous number of semiconductor devices in VLSI circuits ($\sim 10^5$ /chip) virtually demands that the devices themselves have very high yields and that the process for IC fabrication be as simple as possible (as few steps as possible) and have high yield in every step. While the realization of high yield is quite dependent on the state of development of the fabrication technology, certain device approaches clearly are more easily fabricated and promise higher yields in the context of the technology known at a given time. Hence, whereas from a fundamental device point of view, the heterojunction bipolar transistor appears the device of choice for ultra-high-speed, low-power logic, major fabrication technology problems need to be solved before the device can even seriously be evaluated for logic ICs, let alone actually be used for VLSI circuits.

From a practical fabrication/yield standpoint, the only GaAs device approach having demonstrated potential for LSI/VLSI circuits is $L_g = 1$ μm DMESFETs (with 1000-gate circuits demonstrated with Schottky diode-FET logic). While these 1- μm gate length, MESFETs (at $\tau_d \approx 50$ –150 psec) are an order of magnitude slower than what heterojunction bipolar circuits may one day achieve, they exist today and they are still nearly an order of magnitude faster than available silicon homojunction bipolar circuits and take a small fraction of the power. Figure 2 shows the structure for a planar, double-implanted GaAs MESFET, fabricated by localized implantations into semi-insulating ($\sim 10^8 \Omega \text{ cm}$) GaAs [1,2]. The basic structure is very simple, comprising a thin (~ 1000 –2000-Å) n -type (typically $N_d \approx 2 \times 10^{17} \text{ cm}^{-3}$) active region joining two ohmic contacts (separated typically by a distance of $L_{\text{ch}} = 4$ –5 μm) with a narrow ($L_g \approx 1$ μm) metal Schottky barrier gate separating the source and drain. The GaAs FET operation is similar to that of a normal JFET, with the

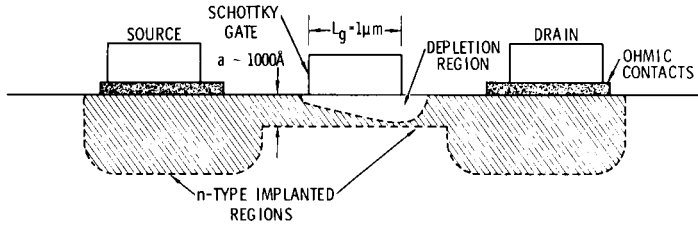


Fig. 2. Cross section of the planar GaAs Schottky gate field-effect transistor (MESFET) structure used in SDFL LSI circuits. The source-drain ohmic contact spacing is $4 \mu\text{m}$ and the deep n^+ contact implant spacing is $2.5 \mu\text{m}$ in this $L_g = 1 \mu\text{m}$ geometry.

conducting n -channel confined between the gate-depletion region and the semi-insulating GaAs substrate. By varying the thickness and doping level of the active region, the pinch-off voltage V_p and saturated drain current I_{dss} (at $V_{gs} = 0$) for the GaAs MESFET can be varied, even down to the point where the normal built-in voltage of the Schottky barrier gate can pinch off the FET (enhancement-mode MESFET operation). Note in Fig. 2 that one narrow ($L_g = 1 \mu\text{m}$) metal line is required, but that the alignment accuracy (approximately $\pm 1 \mu\text{m}$) required is not too stringent for conventional projection or contact photolithographic methods. For this reason, the device fabrication for this $L_g = 1 \mu\text{m}$ GaAs MESFET technology is readily accomplished with commercial step-and-repeat projection aligners, including, presumably, those capable of large production volume levels.

We have reviewed in this section and summarized in Table III what should be achievable with various GaAs device approaches in future ultra-high-speed VLSI. Rather than concentrate further on what could be done with different devices if you could make VLSI complexity circuits of them, we shall shift our focus to whether it is reasonable to expect that circuits of such complexity can be fabricated in GaAs at all. The most practical approach to demonstrating that GaAs VLSI is indeed reasonable is to describe the process approaches used at Rockwell in obtaining our 1000-gate LSI GaAs ICs and yield and performance results obtained from this work.

III. YIELD ANALYSIS AND IMPLICATIONS FOR VLSI DESIGN

The largest fully working GaAs integrated circuits fabricated to date, the Rockwell SDFL 1008-gate, 8×8 -bit, latched parallel multipliers, contain 6000 semiconductor devices (3000 DMESFETs and active loads

and 3000 Schottky barrier diodes), 12,000 circuit overcrossings and 7000 VIA interconnections between the first- and second-level metallizations, on a chip area of 6.7 mm². If we consider even a "lower VLSI" part such as a 1-nsec access time, 4K-bit static RAM, the chip area would be only slightly larger (~8.7 mm² est.), but there could be 60,000 or more semiconductor devices (30,000 FETs and active loads and 30,000 diodes) with nearly 50,000 VIAs and 75,000 overcrossings. To achieve fully working circuit operation, it is necessary that none of these devices or circuit features be fatally defective (e.g., a shorted overcrossing, open VIA, or broken gate on a MESFET). It further requires that all the semiconductor devices (MESFETs and diodes) must be, not only functional, but functional with their key device parameters (e.g., FET pinch-off voltage or diode forward voltage drop) lying within a range that will allow proper circuit functioning (both static and dynamic). These two yield-limiting factors, which we shall refer to as "fatal defects" and "statistical parameter variations," respectively, are best treated independently for clarity in understanding the VLSI yield. In this view, the overall yield Y_{CHIP} for an IC chip is given as the product of two yield factors,

$$Y_{\text{CHIP}} = Y_D Y_S, \quad (9)$$

where Y_D is the yield associated with the finite density of "fatal defects" and Y_S the yield associated with the finite probability that the parameters of one or more of the devices making up the circuit may be out of the range required for proper static or dynamic performance.

A. Yield Reduction from Density of Fatal Defects

It would be possible, in principle, to consider the yield due to such "fatal defects" as overcrossing shorts on the basis of the probability of one overcrossing being good, taken to the power of the number of overcrossings. These probabilities are so close to unity as to lack physical meaning (aside from statistical insight) and device features such as overcrossings (unlike the semiconductor device parameters) have no physical basis for showing an inherent statistical nature that would explain non-unity feature yields. Useful insights into these fatal-defect yield-loss mechanisms have been made on the basis of associating the occurrence of these feature or gross device failures with an assumed finite area density for such defects. This basic approach has proven very useful in explaining silicon LSI and VLSI circuit yields as a function of chip area and chip scaling. In this model, the defect-related chip yield is associated with the probability of including fatal defects in a given chip area. If the density of

these fatal defects is D_D (per square centimeter) and the area of the IC chips is A_C (in square centimeters), then the mean number of defects per chip on the wafer will be

$$\bar{N}_D = A_C D_D. \quad (10)$$

Considerable effort has gone into developing yield models based on various assumptions about the nature of these area-defect distributions (in particular, the degree of correlation between defect locations). The simplest model is to assume that the defects are totally uncorrelated, so that Poisson statistics apply. This model is known to give excessively pessimistic results for LSI chip yields as area is increased, but it will serve adequately to illustrate the point here. Assuming uncorrelated defects, the probability of finding any number (positive integer) N_D of these defects in a chip is given by the Poisson distribution as

$$P_D(N_D) = \exp(-\bar{N}_D) \bar{N}_D^{N_D} / N_D! \quad \text{for } N_D = 0, 1, 2, 3, \dots \quad (11)$$

In most integrated circuit chip architectures, even one fatal defect is too many for the circuit to function. Hence, in this "zero-allowable-defects" IC approach, the chip yield is given by

$$Y_{D0} = P_D(0) = \exp(-\bar{N}_D) = \exp(-A_C D_D). \quad (12)$$

Typical values of defect-density–chip-area products and their associated "zero defect" yields Y_{D0} are shown in the second column, labeled Y_0 , in Table IV. From Eq. (12), we would predict, for example, that if the 1000-gate 8×8 -bit multiplier chip with $A_C = 6.7 \text{ mm}^2$ gave a 20% yield ($D_D = 1.6094/0.067 \text{ cm}^2 = 24/\text{cm}^2$), then a 4K-bit static RAM chip requiring $A_C = 9 \text{ mm}^2$ would give a yield of $\exp(-24 \times 0.09) = 11.5\%$.

In fact, for parts such as memory chips, which are very regular in structure, we need not limit ourselves to accepting the stringent "zero-allowable-defect" criterion for yield. If we can provide some number of redundant rows (or columns) N_{RR} in the memory array to be reprogrammed into use at the wafer probe to replace rows containing defective cells, then we can accept up to N_{RR} , rather than zero, defects on the chip and still obtain functional parts (assuming that all defects are in array itself). For a circuit tolerating up to N_{RR} (rather than 0) defects, we have, for the chip yield,

$$Y_{DN_{RR}} = \sum_{N_D=0}^{N_{RR}} P_D(N_D), \quad (13)$$

where $P_D(N_D)$ is found from Eq. (11). In general, $Y_{DN_{RR}}$ is much higher, of course, than Y_{D0} , as illustrated in Table IV for various defect-density–chip-area products and N_{RR} values from 0 to 6.

TABLE IV

Yield Improvement Effect from Memory Array Redundancy Assuming Poisson Statistics

CHIP AREA x DEFECT DENSITY $\bar{N}_D = A_C \times D_D$	NUMBER OF REDUNDANT ROWS, N_{RR}						
	0	1	2	3	4	5	6
	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6
0.6931	50%	84.66%	96.67%	99.44%	99.92%	99.99%	99.999%
1.6094	20%	52.19%	78.09%	91.99%	97.58%	99.38%	99.86%
2.3026	10%	33.03%	59.54%	79.88%	91.59%	96.99%	99.06%
2.9957	5%	19.98%	42.41%	64.82%	81.60%	91.65%	96.67%
3.9120	2%	9.82%	25.13%	45.08%	64.60%	79.87%	89.83%
4.6052	1%	5.61%	16.21%	32.48%	51.23%	68.49%	81.73%
5.2983	0.5%	3.15%	10.17%	22.56%	38.98%	56.38%	71.74%
6.2146	0.2%	1.44%	5.31%	13.31%	25.74%	41.19%	57.19%
6.9078	0.1%	0.79%	3.18%	8.67%	18.16%	31.26%	46.35%
7.6009	0.05%	0.43%	1.87%	5.53%	12.49%	23.06%	36.45%
8.5172	0.02%	0.19%	0.92%	2.98%	7.36%	14.83%	25.44%
9.2103	0.01%	0.10%	0.53%	1.83%	4.83%	10.35%	18.83%
9.9035	0.005%	0.05%	0.30%	1.11%	3.11%	7.08%	13.63%
10.8198	0.002%	0.02%	0.14%	0.56%	1.70%	4.18%	8.63%
11.5129	0.001%	0.01%	0.08%	0.33%	1.07%	2.75%	5.99%

To illustrate the effect of this redundancy on yield, consider the simple case when the average number of defects per chip is $\bar{N}_D = A_C D_D = 1$. Here the yield of perfect parts is $Y_0 = 36.8\%$, while adding only one redundant row would double it to $Y_1 = 73.6\%$. For $\bar{N}_D = 3$, a higher defect density would drop the yield of perfect parts to $Y_0 = 5\%$, while even one redundant row would quadruple that yield to $Y_1 = 20\%$, and more rows would give even better results ($Y_2 = 42.3\%$, $Y_3 = 64.7\%$, and $Y_4 = 81.5\%$). As shown in Table VI, more dramatic improvements are obtained in cases where the defect density is so high as to be a total "wipeout" for nonredundant parts. For example, with a mean defect count of $\bar{N}_D = A_C D_D = 6.908$, the yield of "perfect" parts would be $Y_0 = 0.1\%$ (unusably low), but only four redundant rows would raise this to $Y_4 = 18.2\%$ and six redundant rows would give a very reasonable $Y_6 = 46.4\%$ from this otherwise useless wafer. Even providing six redundant rows would increase the memory array area by less than 10%. Reprogramming the replacement rows would add significant cost to the test procedure, but if this correction process were reasonably automated, the costs should be more than amply repaid by the increased yield, at least until the "perfect part" yield gets into the $Y_0 = 50\text{--}80\%$ region.

B. Yield Reduction from Statistical Variations of Device Parameters

While the chip-area increase from the 1000-gate, GaAs SDFL, 8×8 -bit, multiplier chip (6.7 mm^2) to that estimated for a 1-nsec, 4K-bit, static RAM ($\sim 8.7 \text{ mm}^2$) is not too large, the increase in the number of GaAs MESFETs (from 3000 to 30,000) and diodes (same tenfold increase) is quite dramatic. Thus VLSI parts with 10,000 + gates would have nearly 10^5 semiconductor devices on each chip. This causes yield problems over and above the "fatal-defect-density" problem discussed in the previous section.

The achievement of ultra-high-speed VLSI requires low-power circuits having very low dynamic switching energies, which is made possible by operation with reduced logic voltage savings ΔV_L (as discussed in Sections I and II). As ΔV_L is reduced, however, the uncertainties in gate or FET threshold voltage may no longer be negligible in comparison to the logic voltage swing. This means, for example, that there is a finite statistical probability that one of the FETs will have a threshold voltage so far from the mean that the logic gate will not function properly (either statically or dynamically) in the circuit. These statistical variations were, in the main, unimportant in older $V_{DD} = 5 \text{ V}$ supply silicon NMOS LSI chips, since the several-volt logic swings were very large in comparison to typical standard deviations in threshold voltage of the MOSFETs. However, as high-performance VLSI with logic swings approaching 1 V emerges, this issue will be important for silicon NMOS, just as it is today in the ultra-high-speed, low-power GaAs circuits. This problem also becomes drastically worse as the number of devices per chip is increased. In fact, it might best be called the "stone wall" effect because of the sudden, drastic dropoff of yield in parts with large numbers of devices when the ratio X of the allowable limit for a parameter ΔV_A (measured from the mean) to the standard deviation of that parameter σ_V ,

$$X = (V_{UL} - \bar{V})/\sigma_V = \Delta V_A/\sigma_V, \quad (14)$$

gets too large. If we assume Gaussian statistics for the device parameter, then the probability that a given device will be beyond the acceptable excursion limit of ΔV_A from the mean will be $P(\Delta V_A/\sigma_V) = P(X)$, where $P(X)$ is the well-known error integral [12]. If this is the only statistical parameter limiting yield and there are N such devices with the same limit V_A and statistics σ_V on the chip, then the statistical chip yield Y_S in Eq. (9) will be given as

$$Y_S = [P(X)]^N \quad (\text{single-limit case}). \quad (15)$$

This function is graphed in Fig. 3 for various numbers N of devices on a

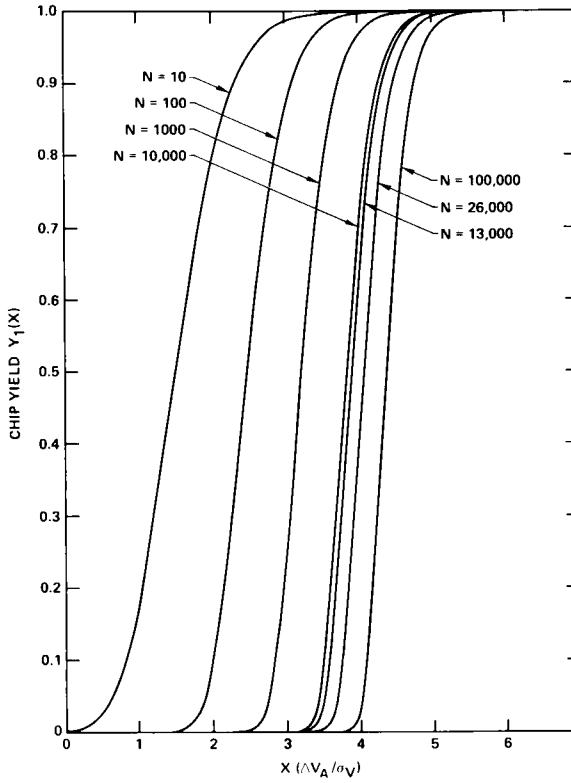


Fig. 3. Chip yield due to statistical parameter variations, calculated assuming Gaussian statistical variations of a parameter (like FET V_p) and a single (e.g., upper) limit on the value of that parameter for circuit operability. Yield Y_s is plotted for various numbers N of devices in the chip as a function of X , the ratio of the difference between the limit and the mean ΔV_A and the standard deviation σ_v using Eqs. (14) and (15).

chip. Note that for $N = 13,000$, taking a parameter limit from $\Delta V_A = 4.5\sigma_v$ down to $\Delta V_A = 3.5\sigma_v$ drops the yield from over 95% to under 5%, and going to $\Delta V_A = 3\sigma_v$ reduces the yield to 2.4×10^{-8} . Actually, in general, there are both upper and lower limits on a parameter, typically with, for best yield, the nominal mean located in the center, so that the yield expression for Eq. (9) should be

$$Y_s = [P(X^+)P(X^-)]^N \quad (\text{two-limit case}), \quad (16)$$

where $X^+ = (V_{UL} - \bar{V})/\sigma_v$ and $X^- = (\bar{V} - V_{LL})/\sigma_v$, V_{UL} the upper limit of the parameter, V_{LL} the lower limit, and $\bar{V}$ the mean of the parameter over the wafer. For the simple example of an optimum wafer mean, $\bar{V} = \frac{1}{2}(V_{UL} + V_{LL})$. Then taking $\Delta V = \frac{1}{2}(V_{UL} - V_{LL})$, Eq. (15) and Fig. 3

still apply except that we must replace N by $2N$. (That is, a 13,000-device chip would use the $N = 26,000$ curve in Fig. 3; this is even steeper, dropping from 91.5% at $4.5\sigma_v$ to 0.24% at $3.5\sigma_v$.)

In real integrated circuits, the situation is even worse than illustrated in Fig. 3 because of the fact that the *mean itself* $\bar{V}_p$ varies from wafer to wafer, usually with a standard deviation σ_w exceeding that of the parameter variation over a given wafer (σ_v). This distribution of means $\bar{V}_p$ for an assemblage of wafers has a degrading effect on average yield, which can be calculated by integrating Eq. (16) over the distribution of wafer means with the appropriate normalized probability weighting to give the average. A computer program for this analysis has been written for the case when the "mean of means" is optimum [$\bar{V}_p = \frac{1}{2}(V_{UL} + V_{LL})$] and the distribution of $\bar{V}_p$ is Gaussian with a standard deviation of σ_w . For example, if we consider an EMESFET RAM with a pinch-off voltage limit separation of $V_{UL} - V_{LL} = 300$ mV for circuit function ($\Delta V_A = 150$ mV), then we find, if we take (a very optimistic value of) $\sigma_v = 35$ mV for the standard deviation of pinch-off voltage, from Fig. 15 (with $X = 4.286$ on the $N = 26,000$ curve), a theoretical maximum yield for $N = 13,000$ FET RAMs of 78.9% for a wafer with precisely optimum $\bar{V}_p$. Among a group of wafers with an average value of $\bar{V}_p$ at optimum but with a standard deviation $\sigma_w = 70$ mV (also very optimistic), the average theoretical maximum yield would fall to less than 15%, while for a more realistic $\sigma_w = 150$ mV, to less than 7%. In fact, a very good standard deviation of GaAs MESFET pinch-offs over one wafer is $\sigma_v = 45$ mV (from some of our planar IC wafers), which would give, for $V_{UL} - V_{LL} = 0.3$ V, a theoretical maximum yield of 1.4×10^{-5} with $\sigma_w = 0$ (5.5×10^{-7} with $\sigma_w = 100$ mV).

This example can also serve to illustrate the extreme danger of extrapolating SSI yield data to VLSI parts. While the $\sigma_v = 45$ mV EMESFET example for the $N = 13,000$ FET RAM gave an unusable $\bar{Y}_S = 5.5 \times 10^{-7}$ with $\sigma_w = 100$ mV, this same wafer collection would give an average yield of $\bar{Y}_S = 65\%$ on an $N = 5$ FET part or $\bar{Y}_S = 18\%$ for an $N = 10$ FET SSI chip. Even if σ_v and σ_w were both as high as 100 mV, $N = 10$ FET SSI parts would yield up to 10.5% (or 27% for $N = 5$), so that it would be easy to feel, on the basis of SSI results, that FET pinch-off control was "no problem," when in fact yield would hit the "stone wall" at $N = 30$ ($\bar{Y}_S = 0.09\%$). Yield for $N = 13,000$ FETs would be, of course, nonexistent.

The conclusion to be drawn from this examination of the relation of statistical parameter variations to yield is that it would be *extremely* unwise to make a commitment to a device technology approach for VLSI, whatever the apparent attractions, if the peak-to-peak parameter variation limits ($2V_A$) allowable in the circuit approach (for proper static and

dynamic operation) are not at least ten times the standard deviation for the device parameter (e.g., σ_{V_p}) reasonably expected to be obtained with the fabrication process. Typically, this will mean that the logic voltage swing ΔV_L must be at least 20 (preferably more) times the standard deviation of FET pinch-off voltage in FET logic to avoid dynamic switching speed and noise-margin degradation. With the present "good" $\sigma_{V_p,s} \approx 50$ mV for the best MESFET process and the logic swings of GaAs EMESFET logic limited to $\Delta V_L \approx 500$ mV, it would appear that this technology is not quite ready to attack VLSI. (The somewhat larger voltage swings allowable in EJFET circuits would, for the same $\sigma_{V_p,s}$, help this problem).

In our depletion-mode SDFL logic circuits (see Fig. 4a), the typical voltage noise margins are of the order of 0.6 V or more on either a "high" or "low." In comparison to the observed $\sigma_{V_p} \approx 50$ –100 mV wafer uniformities (let alone the $\sigma_{V_p} \approx 17$ –30 mV short-range uniformities), the statistical probability of exceeding the voltage noise margin is negligible. This is a consequence, of course, of the fact that the logic voltage swings and noise margins have been kept fairly large in the SDFL circuit designs in order to cope with the significant switching transient-induced ("hash") noise levels that are commonly present in very high speed digital circuits and systems. The *current* noise margins, or relationship between the current drive capability of a gate and the input current requirements of its fanout of loading gates, is a different issue, however. In general, optimum switching-speed performance is obtained when the source-drive capabilities and loading-gate input-current requirements are closely matched. This is always the case for transient currents in dynamic measurements on FET logic circuits, but it is even more obvious in SDFL, where this is also reflected in a static fanout limitation because of the dc input requirements of the gates (Fig. 4a). The static fanout and, hence, the current noise margin, may be increased in the SDFL circuits by increasing the ratio of the output pullup (PU) to the input pulldown (PD) active load current (by adjusting device widths). On the other hand, increasing this ratio much beyond that required to meet the particular fanout loading conditions for the gate involves some sacrifice in switching speed. For this reason, most of our SDFL circuits are fabricated with a current noise margin of only about 30%. With the observed long-range standard deviations in device saturation currents $\sigma_{I_{DSS}}$ in the 6–8% range, little influence on yield would be expected, even for >1000-gate LSI circuits. If $\sigma_{I_{DSS}}$ were 10%, significant yield reductions (to about 26% on 1000-gate circuits) would be expected, so that it would be necessary to raise the current noise margin slightly. For VLSI designs, it will be important to watch these active load statistics carefully and to optimize the perform-

ance/yield trade-off according to the statistical device parameter control available in the process.

IV. THE SCHOTTKY DIODE-FET LOGIC CIRCUIT APPROACH FOR ULTRA-HIGH-SPEED GaAs VLSI

The achievement of LSI or VLSI requires high gate densities, as attained in part in this work by the use of high-resolution ($1\text{-}\mu\text{m}$) lithographic techniques. Another important factor defining the number of logic gates attainable per unit chip area is the circuit design used. Whereas in all previously published GaAs FET digital IC work the GaAs FETs themselves were used as the nonlinear logic elements, with this planar, multiple-localized implantation approach, optimized GaAs Schottky barrier switching diodes can be fabricated. Inasmuch as GaAs Schottky barrier diodes are very strongly nonlinear, are among the fastest-switching semiconductor devices existent, and have very low switching energies ($\sim 10^{-15}$ J), required chip areas ($1 \times 2\text{-}\mu\text{m}$ active area), and capacitances (~ 2 fF), their use as logic elements in GaAs digital ICs seems very desirable. We have developed a logic circuit approach using such high-speed GaAs Schottky barrier switching diodes [13,14] for most logic functions with, in its simplest form, GaAs Schottky gate FETs used primarily for inversion and gain, as shown in Fig. 4a.

Figure 4a shows the circuit diagram for a simple four-input Schottky diode-FET logic NOR gate. The logical OR function (prior to inversion) is accomplished by the very small high-conductance GaAs Schottky barrier diodes. In addition to their logic function, these diodes provide ~ 0.75 V of level shifting between the positive logic levels from previous stages and the negative gate voltages required by the normally "on," depletion mode Schottky gate GaAs FETs. For every low pinch-off voltage GaAs FETs ($V_p \approx -0.5$ V), the switching diodes ($D_A - D_D$) alone provide adequate level shifting for very low voltage and very low power ICs. By adding the level shift diode D_s in Fig. 4, GaAs FETs with slightly high pinch-offs ($V_p \approx -0.7$ to -1.4 V) may be used with higher supply voltages and logic swings; the data presented here are for gates with this shift diode.

The pulldown PD to $-V_{ss}$ sinks the bias current for the diodes and provides most of the turn-off current for the gate of Q1. The pulldown is kept small (typically $\leq 20\%$ of the width of Q1). However, since in this current-sourcing logic the dc fanout is limited to the current ratio between PU and PD (typically 3 or 4), larger fanouts are achieved, where necessary, by adding a source-follower or other output buffer configuration to

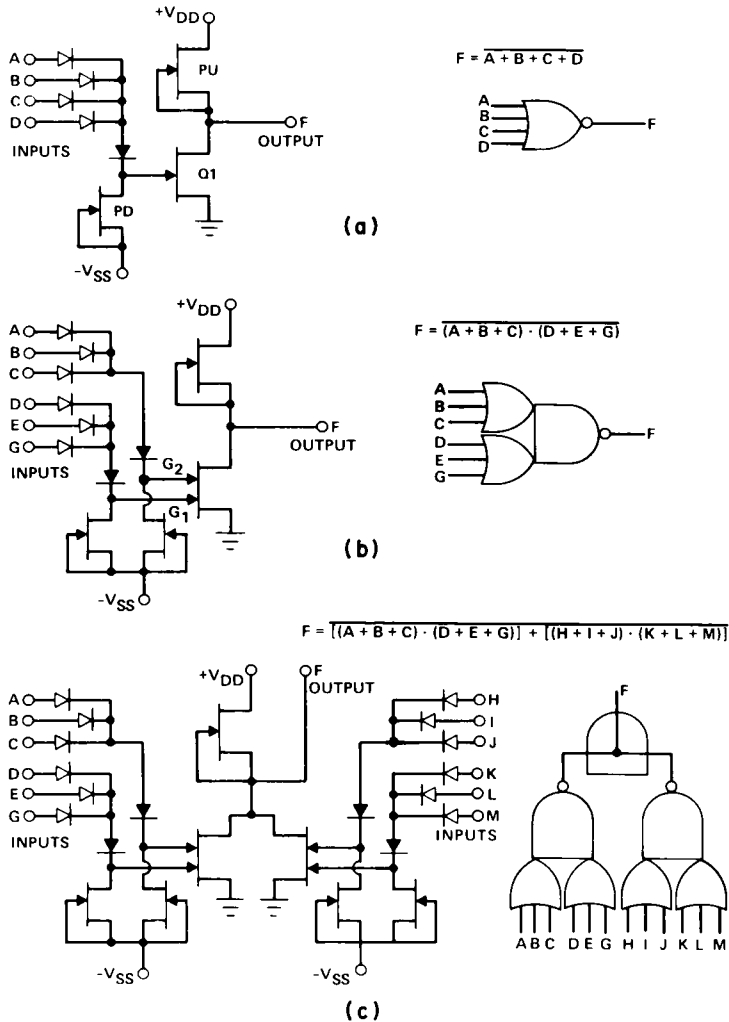


Fig. 4. Schottky diode-FET logic (SDFL) gate configurations for one-, two- and three-level logic gates: (a) SDFL NOR gate; (b) SDFL OR/NAND gate; (c) SDFL OR/NAND/WIRED-AND gate. All FETs are depletion mode, typically $-1.5 \text{ V} < V_p < -0.5 \text{ V}$; unshaded diodes are very small high-speed switching Schottky diodes, while shaded diodes are larger-area, higher-capacitance voltage-shifting diodes.

the gate. Because of the very small size of the logic diodes and because of the fact that the diodes, which are two-terminal devices, require many fewer overcrossings and VIAs than circuits using three-terminal devices like FETs, multiinput SDFL NOR gates require only a fraction of the chip area needed by FET logic approaches. This results in lower parasitic

drain and stray capacitances for multi-input SDFL NOR gates, for improved dynamic switching performance in comparison to multi-input gates in FET logic.

In the interests of efficiency and speed, it is often desirable to implement logic functions with multilevel gates. In FET logic, the use of combinations of series (NAND) or parallel (NOR) FET configurations to achieve two-level logic functions is common. For example, in earlier GaAs work, Van Tuyl *et al.* [15] demonstrated depletion-mode GaAs MESFET buffered-FET logic (BFL) NAND/NOR (or NAND/WIRED-AND) gates, with up to two NAND input terms (series or dual-gate FETs) and up to two of these NAND functions drain-dotted together, having propagation delays as low as $\tau_d = 110$ psec. Four of these gates were utilized to implement a fast [$f_c \approx 1/(2\tau_d)$] complementary-clocked $\div 2$ frequency-divider stage, which gave toggle frequencies up to 4.5 GHz for $\tau_d \approx 111$ psec effective gate delays at a $P_D \approx 40$ mW/gate power level [15]. Here, the advantage of a two-level gate is clear, since the fastest NOR-gate-implemented complementary-clocked divider circuit uses eight gates and has a maximum speed of $f_c = 1/(4\tau_d)$. In SDFL, we have fabricated more easily used, single-clocked T-connected D-flip-flops using six NOR gates, achieving maximum toggle frequencies of $f_c = 1.9$ GHz at $P_D = 2.5$ mW/gate, which corresponds to $\tau_d = 110$ psec ($f_c = 1/(4.85\tau_d)$) in this configuration [1,2,16]. Hence, while the SDFL NOR gates gave the same propagation delays as the BFL gates (and at much lower power levels), the architecture of the NAND/NOR dividers allowed much higher toggle frequencies to be reached; this has been (inaccurately) cited as a significant limitation of the SDFL circuit approach.

In fact, such multilevel logic gate configurations may indeed be realized in SDFL [17] with up to three-level gates with many (10–20) inputs, not being restricted to four-input two-level gates as in BFL. Thus, the Schottky diode–FET logic subsumes normal FET logic, to which is added the additional level of logic provided by the ultra-high-speed switching diodes.

In our earlier work [13,14,18], the FET logic function utilized was principally the inverter, so that, with the diode OR, a NOR gate function was realized (Fig. 4a). On the other hand, by using such diode OR clusters on each gate of a dual-gate FET (or series FET connection), an OR/NAND two-level gate is achieved (Fig. 4b), which is the complement of the two-level NAND/NOR gate realized in BFL, except that the number of first-level terms in the SDFL version is no longer restricted to two. Furthermore, drain dotting can be used between two of these SDFL OR/NAND gates to achieve a three-level OR/NAND/WIRED-AND gate configuration (Fig. 4c), achieving the (positive) logic function

$$F = [(A + B + C) \cdot (D + E + G)] + [(H + I + J) \cdot (K + L + M)] \quad (17)$$

or, equivalently,

$$F = [(A + B + C) \cdot (D + E + G)] \cdot [(H + I + J) \cdot (K + L + M)] \quad (18)$$

While the application of the two-level SDFL OR/NAND gates of Fig. 4b is obvious because of its complementary relationship to the widely used NAND/NOR configuration [15], the utility of the three-level SDFL OR/NAND/WIRED-AND gates (Fig. 4c) is less clear and best illustrated with an example. One of the major accomplishments of our GaAs IC effort has been the demonstration of a 1008- (NOR) gate, 8×8 -bit, latched parallel multiplier. The parallel multiplier array in this part consists of 760 NOR gates, principally used in the 48 full-adder cells that (with eight half-adders) make up the array. A 12-gate, $3\tau_d$ -to-sum and $2\tau_d$ -to-carry, full-adder cell configuration was selected for this design. The speed of the

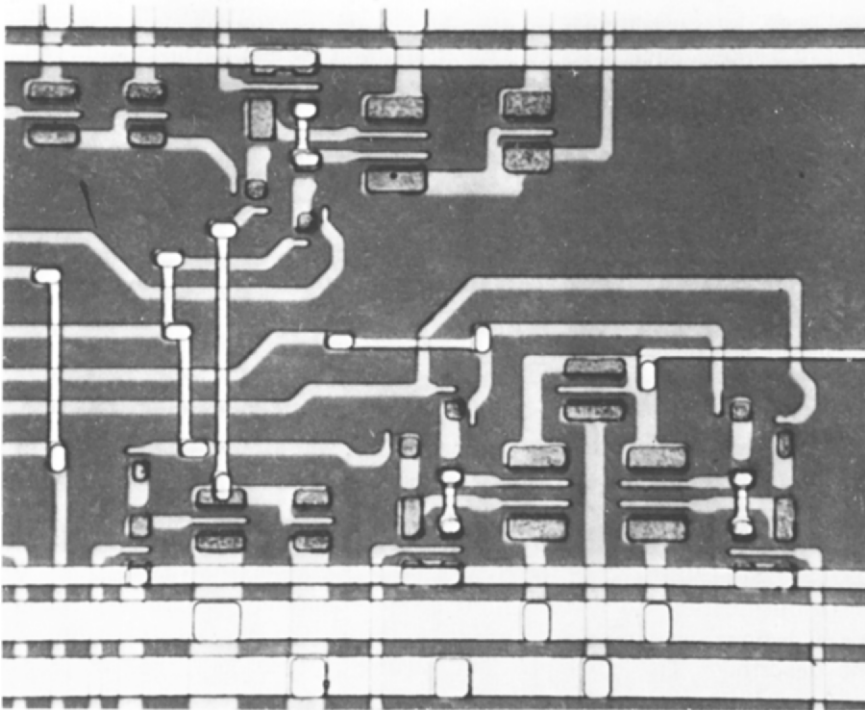


Fig. 5. Microscope photograph of a portion of a planar SDFL circuit (Fig. 4 of Ref. 15) showing one-, two- and three-level SDFL gates. At the lower left is an inverter (one-input NOR gate), at the top center a two-level OR/NAND gate, and at the lower right a three-level OR/NAND/WIRED-AND gate.

multiplier could be improved by reducing the number of logic gate propagation delays τ_d required to accomplish the full-adder function. Even with NOR gates a $2\tau_d$ delay for both sum and carry can be achieved, but at the (prohibitive) expense of an 18-gate cell complexity. By using two-level and three level SDFL gates, a full-adder circuit (Fig. 3 of Eden *et al.* [17]) can achieve a $1\tau_d$ -to-carry (C or $\overline{C}$) and $2\tau_d$ -to-sum (S or $\overline{S}$) delay with only six gates (four OR/NAND and two OR/NAND/WIRED-AND). This would reduce the delay for an 8×8 -bit parallel multiply from the $35\tau_d$ of the NOR implementation to $21\tau_d$, a 40% reduction. The power should be greatly reduced since only half as many gates are required, although the yield would not be expected to be improved since similar numbers of logic FETs are required. The essential speed advantage of the three-level OR/NAND/WIRED-AND gate in generating $\overline{C}_{OUT}$ from C or C_{OUT} from $\overline{C}$ in only $1\tau_d$ can be utilized in fast ripple-adders without generating all four (S , $\overline{S}$, C_{OUT} , and $\overline{C}_{OUT}$) terms by changing from propagating C_{OUT} to $\overline{C}_{OUT}$ in alternate stages. This approach (illustrated in Fig. 4 of Eden *et al.* [17]) has been demonstrated in a fast ripple-carry 4-bit adder chip (Table I). Figure 5 shows a microscope photograph of a section of this adder circuit showing one-, two-, and three-level SDFL gates.

V. PLANAR GaAs LSI/VLSI FABRICATION TECHNOLOGY

During the past several years, the development of GaAs integrated circuits have been making rapid progress [1]. Currently, several GaAs IC technologies have achieved MSI status and are expected to impact near-term high-speed signal-processing applications [2]. However, as the trend of modern electronics suggests, for GaAs IC technology to make a serious impact on future electronic systems, it must be capable of LSI/VLSI complexities. Gallium arsenide fabrication approaches and processing techniques must be selected that are extendible and fully compatible with the yield and reliability demands of high-density, complex LSI/VLSI [19].

In this section Rockwell's fabrication technology, which was specifically designed and developed to satisfy the goal of LSI/VLSI [19,20], will be described. This goal provided strong motivation for developing a planar fabrication process using the low-power Schottky diode-FET logic (SDFL) circuit approach shown in Fig. 4. In the following we shall discuss a planar GaAs integrated circuit fabrication technology, which has been demonstrated at the LSI level and is potentially capable of achieving VLSI. This technology combines advanced planar device and multilevel

interconnect structures with state-of-the-art LSI processing techniques including multiple localized ion implantations, reduction photolithography, plasma etching, reactive ion etching, and ion milling. Using this process technology, LSI-level (1008-gate) GaAs ICs have been successfully fabricated and tested.

A. Planar Ion-Implanted Fabrication Process

The development of an ion implantation technology in GaAs has made it possible to conceive new fabrication approaches not viable within the limitations of epitaxial layers. These GaAs implantation advancements, coupled with the observation that the rapid development of Si LSI was greatly facilitated by planar fabrication techniques using ion implantation and dielectric passivation, suggested that similar developments in GaAs would enhance the development of LSI/VLSI. The schematic of Fig. 6 represents a cutaway view of the Rockwell planar IC approach. The planar circuits are fabricated by using multiple localized ion implants directly into semi-insulating GaAs substrates. Hence, individual devices can be optimized by using different implants, with the unimplanted GaAs substrate providing isolation between adjacent devices. This fabrication method ideally complements the Schottky diode-FET logic circuit approach (shown in Fig. 4) since SDFL requires the MESFETs and Schottky diodes to have different implanted active layers. The sophistication of this process technology is illustrated (Fig. 6) by the use of a

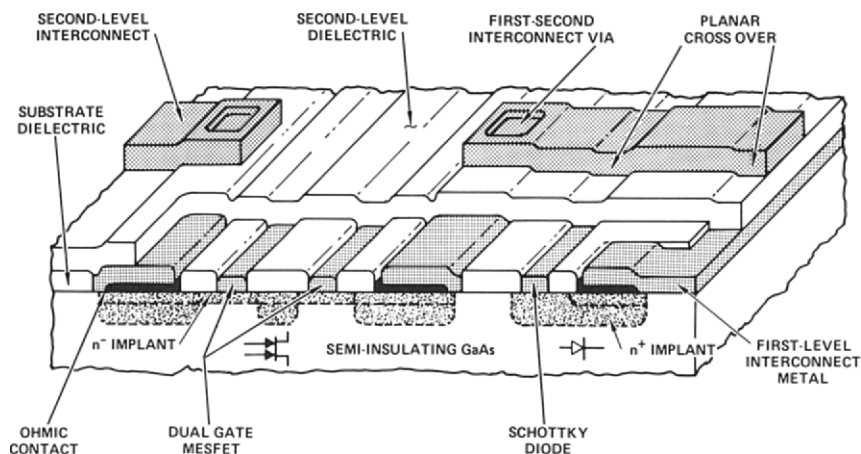


Fig. 6. Cutaway view of a planar GaAs IC showing a dual-gate FET, a diode, and interconnects.

1- μm -wide n^+ implant region placed between dual 1- μm MESFET gates in order to lower the channel resistance between the dual gates. Another important aspect of this approach is the use of dielectrics. Dielectric regions are utilized for post-implantation annealing, protecting the GaAs surfaces during processing, and passivating the surface. This fabrication approach is compatible with any number of implantation steps; therefore, this process technology has, in principle, the capabilities for producing mixed device types, such as EMESFET (or EJFET) and DMESFET on the same GaAs IC chip.

The planar implantation and lithography process steps used in this work are shown in Fig. 7. It is a totally dry (no wet chemical process steps), simple (only six mask levels), high-yield LSI/VLSI-compatible process.

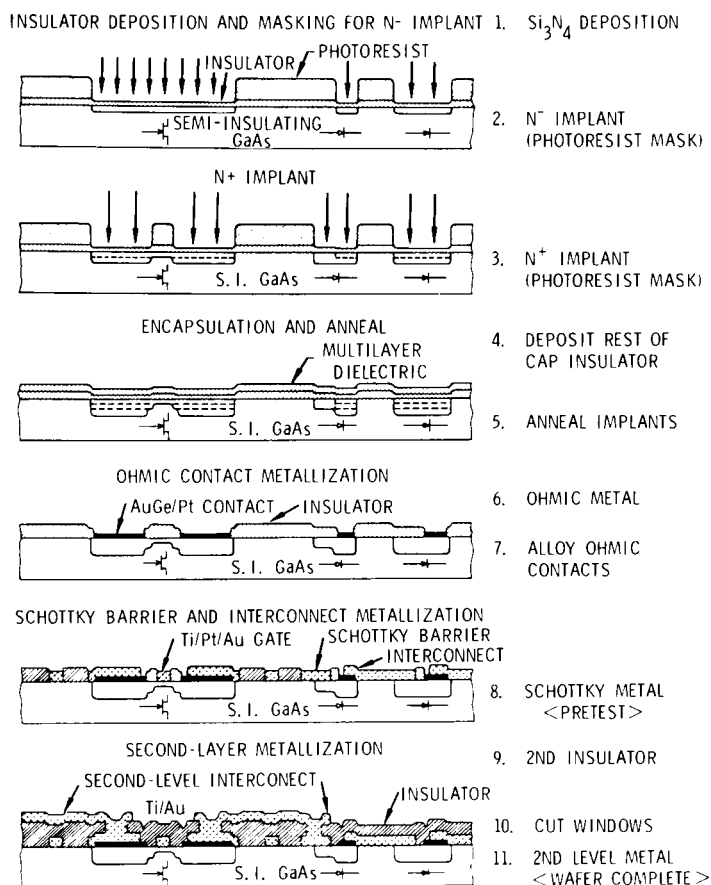


Fig. 7. Planar SDFL GaAs IC fabrication steps.

The material processes are characterized by multiple localized implantations (nonepitaxial), providing planar device structures isolated by high-resistivity semi-insulating GaAs substrates. The use of any number of localized Se, S, or Si implants or combinations of implants provides the process flexibility for optimizing various LSI/VLSI circuits. All the dielectrics and metals are deposited or grown using conventional processing methods including rf diode and magnetron sputtering, e-beam evaporation, and plasma-enhanced CVD techniques.

Resist delineation is accomplished by using reduction–projection photolithography for defect-free, micron-resolved photoresist patterns. Pattern replication is accomplished without any chemical etching. A unique high-yield replication process has been developed by Rockwell specifically for planar GaAs ICs (see Section V.B). This fabrication process incorporates plasma etching in conjunction with an intermediate lift-off technique providing extremely high yield, self-aligned, metal/dielectric patterns. Second-level metal interconnects are replicated using a dry etched, ion-milling technique. As will be discussed in Section V.C, the resulting planar structures are used to great advantage in the fabrication of high-yield, complex, multilevel GaAs ICs.

The implant process steps used in this work are shown in Fig. 7, steps 2–5. Initially, the GaAs is coated with a thin (~ 1000 Å) layer of Si_3N_4 , which remains on the wafer throughout all of the subsequent processing steps. The first process steps are the two localized implantations carried out through the thin Si_3N_4 layer using thick photoresist as the ion-beam mask. Following the implants, additional dielectric (SiO_2) is added prior to the post-implantation annealing shown in step 5.

Recently, the continued development of planar localized implant processes for LSI performance and yield optimization has led to the investigation of Si as an alternate ion implant species for both the Se and S implants normally used in our planar GaAs process [20]. Silicon appears to be a suitable ion species for use as a “universal” n -type implant ion in GaAs. The insert table and profiles shown in Fig. 8 illustrate how Si-implanted profiles can satisfy both the DMESFET and Schottky switching diode requirements.

Typical implantation parameters and the resulting electron concentration doping profiles for both IC implants conducted through the Si_3N_4 cap are shown in Fig. 8. The shallow 400-keV Se or 160-keV Si implantation profiles are peaked near the GaAs surface due to the energy absorbed in penetrating the thin Si_3N_4 layer. In contrast, the S or Si (n^+) implants provide much deeper (~ 4000 -Å) active layers than the shallow (~ 1500 -Å), low-threshold ($V_p \approx -1$ V), MESFET channel (n^-) implant. These deeper-implanted layers are ideally suited for the high-speed switching

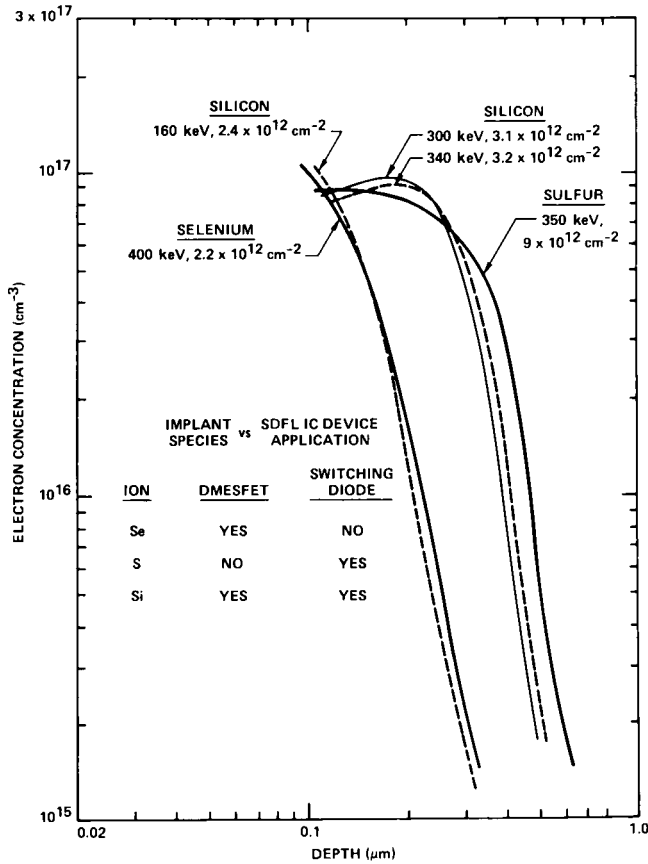


Fig. 8. Carrier concentration profiles for Se, Si, and S implants conducted through the 925Å Si₃N₄ layer.

diode required in SDFL. Both n^- and n^+ implants are used for the level-shifting diodes and for enhancing the doping under ohmic contact regions. As demonstrated in Fig. 8, the appropriate choice of Si implant energy and dose results in profiles similar to the typical Se and S profiles normally used. However, merely substituting Se and S implants with Si implants is not the important issue. In the case of DMESFET channel profiles and, to a lesser extent, Schottky diode profiles, the reproducibility, uniformity, and IC device performance are the real issues relevant to the optimal implant species experimentation. Silicon active layer profile reproducibility and uniformity investigations have been initially carried out by comparing the dc performance of DMESFETs and Schottky diodes fabricated with various Si, Se, and S variations.

The emphasis of implant species comparison experiments was placed on evaluating the shallow, lightly doped, high-resistance, n^- MESFET channel layer. This active layer directly controls the pinch-off voltage V_p , the key parameter in the optimization of the power-delay product of the logic gates. The uniformity of MESFET pinch-off voltages for four implant species combinations is shown in Fig. 9. As shown in Fig. 8, the n^- FET channel is implanted with either Se or SI, while the n^+ ohmic contact

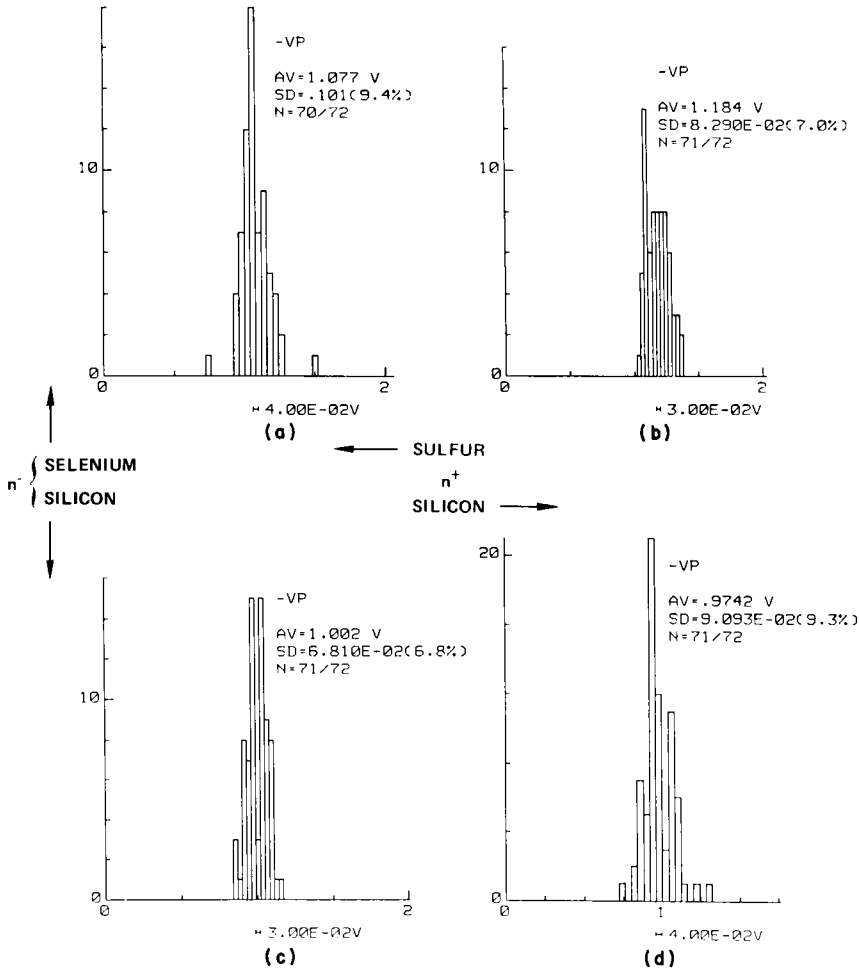


Fig. 9. Histograms of pinch-off voltages of test FETs comparing various implant species combinations of n^- and n^+ implanted regions: (a) (Se,S); (b) (Se,Si); (c) (Si,S); and (d) (Si,Si).

region is implanted with either S or Si. In Fig. 9, the average pinch-off voltage $\bar{V}_p$, measured for 72 FETs equally distributed across the wafer, lies within a narrow 0.974–1.184 range, indicating adequate control for all the dopant variations studied. The uniformity of V_p is also quite good, as indicated by the standard deviation values, ranging from a low of 68 mV (6.8%) to a high of 101 mV (9.4%). The results shown in Fig. 9 are average threshold voltage values (85 mV, 8.0%) routinely obtained with the implant processes described. This implant optimization study has not yet singled out any obvious advantage for choosing one species combination over another. To date, these data support the conclusion that Se, Si, and S are all viable GaAs *n*-type implant dopants when utilized in conjunction with mature material, ion implantation, and encapsulation processes.

Recently, we have experienced 60-mV pinch-off voltage standard deviations for full wafers averaged over many GaAs IC wafer lots. The constant uniformity improvement that we have observed with time is indicative of the advancements made in both the semi-insulating GaAs substrates and the fabrication processes. State-of-the-art results for DMESFET threshold uniformities are shown in Fig. 10 for both long-range (full-wafer) and short-range (FET-array) distributions [7]. Full-

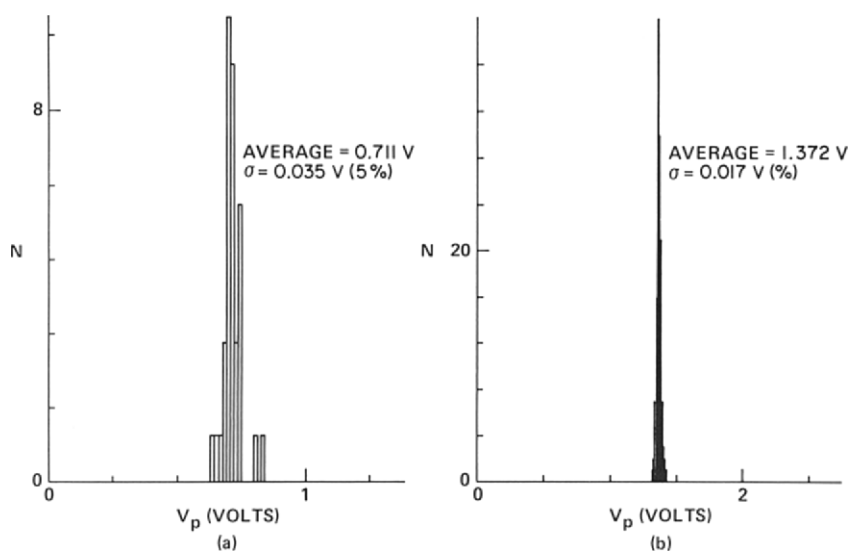


Fig. 10. Histogram of pinch-off voltages of test FETs comparing the uniformity for MESFETs uniformly distributed (a) over an entire wafer area (long range) with that for (b) 81 MESFETs packed in the area of a small chip ($\sim 300 \times 300 \mu\text{m}$; short range).

wafer pinch-off voltage standard deviations of 35 mV have been monitored on numerous wafers with the best short-range data being 17 mV (1.2%). The constantly improving wafer-device uniformity strongly supports the viability of this technology for reaching VLSI levels.

B. Advanced Microstructure Pattern Replication Techniques

First-level circuit lithography processes encompass the most crucial steps in the fabrication of planar GaAs ICs. Since GaAs ICs are designed with 1- μm features to achieve speed goals, a dry pattern replication process capable of extremely high LSI/VLSI-compatible yields has been developed.

The following will summarize the first-level circuit lithography steps used to define ohmic contacts, Schottky barriers, gates, and first-level interconnects. At this point in the process the entire surface of the GaAs wafer is covered with dielectric. Subsequent ohmic contact and Schottky metallization layers are deposited within windows in this dielectric (as shown in Fig. 7, steps 6–8). This process represents a significant departure from typical microwave GaAs MESFET fabrication approaches and more closely resembles Si planar IC processing.

Common to all of the GaAs IC process steps is the delineation of the fine-line resist patterns required for the fabrication of these circuits. Since 1977, photolithography steps for our GaAs ICs have been accomplished by using a Canon FPA 141 4 $\times$ projection mask aligner. Circuit pattern replication is accomplished using projection photolithography in conjunction with photoresist lift-off techniques.

Photoresist lift-off techniques have been commonly used in fine-line ($\sim 1\text{-}\mu\text{m}$) lithography applications for many years. However, when direct photoresist lift-off techniques are used for defining metal patterns, excellent edge acuity of the photoresist profile is required. Normally, in practice, excellent edge acuity (vertical side walls) is not always attainable and limited process yield is commonly experienced with direct photoresist lift-off. This is particularly true when using reduction–projection photolithography techniques, where resist images and profiles are often difficult to control because of the small depth of focus (typically $\pm 1.5\text{ }\mu\text{m}$ for 1- μm resolution) of this optical technique combined with the large wafer flatness variations normally experienced. Therefore, in order to ensure high yields from lift-off processes, enhanced lift-off techniques are commonly used [21]. Two examples of enhanced lift-off techniques are shown in Fig. 11a–f with the plasma-etched, dielectric, intermediate-layer method used in this work shown in Fig. 11g–i. The key principle

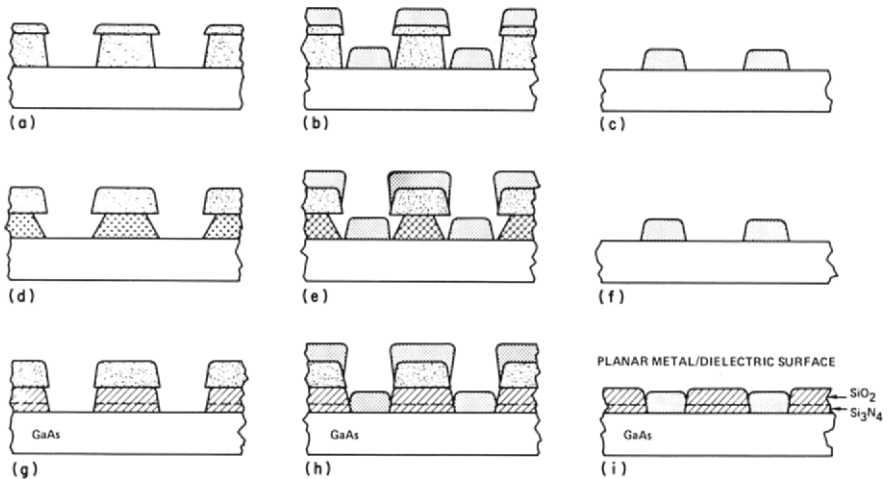


Fig. 11. Schematics of two commonly used enhanced lift-off methods: (a)–(c) the multilevel mask and (d)–(f) the intermediate-layer approaches, in contrast with the plasma-etched dielectric intermediate-layer technique described in this work, using (g) the mask structure, (h) evaporation, and (i) replicated structure.

used in both the multilevel mask or intermediate layer methods is the use of evaporation shadowing. Mask structures are formed with resist ledges (multimask techniques) or under-cut resist/dielectric structures (intermediate layer) that allow convenient shadowing of the evaporated metal, thus ensuring an extremely high yield lift-off process.

The enhanced lift-off method described here was developed specifically for GaAs IC applications. This process approach has taken into account the inherent differences between processing on GaAs versus Si. For example, in this work, the substrate is semi-insulating (the Si analogy would be insulating sapphire substrates in SOS technology); therefore, the first-level interconnects may be fabricated directly on the substrate without the usual constraints associated with isolation and stray capacitance common to Si. Additionally, the commonly used freon-based plasma-etching techniques used for etching Si_3N_4 and SiO_2 do not etch GaAs. These two unique features of GaAs have motivated the development of the plasma-etched, dielectric, intermediate-layer, enhanced lift-off technique shown in Fig. 11. The structural advantage of the intermediate-layer, enhanced lift-off method can be seen in the scanning electron micrograph of Fig. 12a. The intermediate dielectric layer is first plasma etched; then the appropriate metal is evaporated to a thickness less than or equal to the dielectric thickness. The resulting microstructure automat-

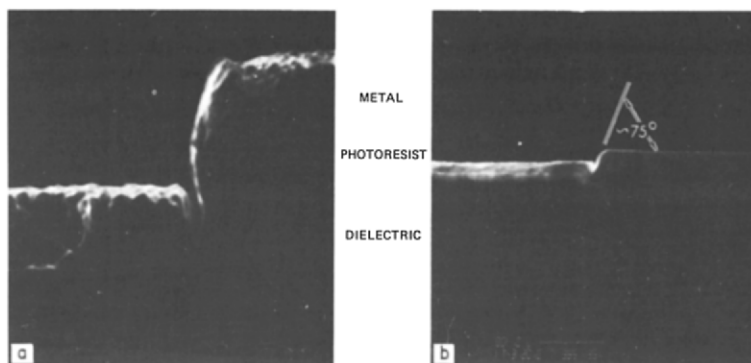


Fig. 12. (a) Scanning electron micrograph showing the intermediate-layer enhanced lift-off microstructure prior to lift-off. (b) Scanning electron micrograph showing the planar self-aligned metal/dielectric structure resulting from the use of the enhanced lift-off method described in this chapter.

ically provides a separation between the metal deposited in the dielectric window and the metal deposited on the top and side wall of the photoresist pattern. This structure allows solvents to easily penetrate and readily dissolve the photoresist. This process is easy to implement, very reproducible, and extremely high yielding.

The key difference between this approach and other intermediate layer lift-off approaches is that plasma-etching techniques have been utilized for replicating micron-resolved dielectric windows with minimum resist degradation, which results in a lift-off process for providing the precisely aligned metal/dielectric structure shown in Fig. 12b. Optimization of the plasma-etching parameters for near-anisotropic ($\sim 75^\circ$ slope) dielectric profiles allows the metal to actually seal the dielectric opening (see Fig. 12b). This is advantageous for GaAs IC processing because critical, lightly doped, thin GaAs MESFET channel active layers are sealed and isolated from subsequent process steps.

This unique, first-level lithography, pattern replication technique has continuously and successfully been used in the development of GaAs LSI.

C. High-Yield Planar Multilevel Interconnects

The Rockwell multilevel interconnect fabrication approach is illustrated in the planar multilevel GaAs IC structure shown in Fig. 13. As illustrated in Fig. 13a, this fabrication approach hinges on maintaining a smooth metal/dielectric surface after the first-level metallization steps,

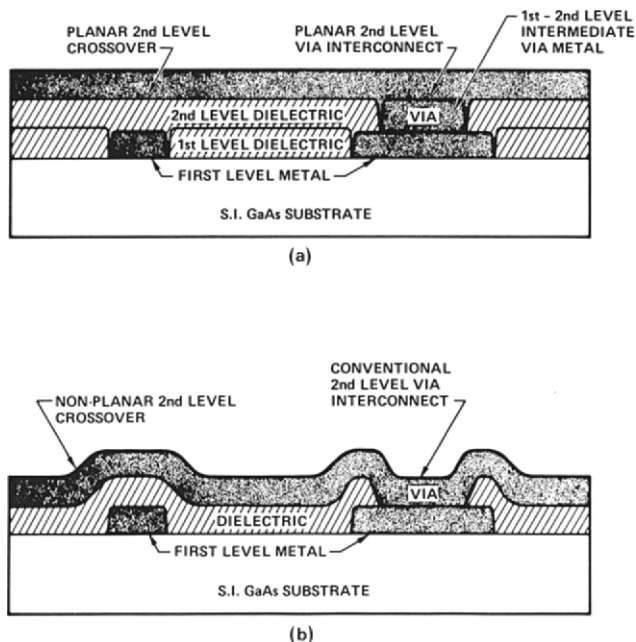


Fig. 13. Schematic of the Rockwell planar multilevel interconnect structure (a) compared to conventional fabrication structures (b).

leading to the planar multilevel interconnect crossover structure shown. Planar crossovers greatly enhance the yield of multilevel interconnects as compared to conventional crossover structures shown in the Fig. 13b. Also shown in Fig. 13 is the use of “filled” VIAs interconnecting the first-level metal to the second-level metal. Since there is no step coverage consideration for filled VIAs, higher-yield, denser, multilevel interconnect structures are possible.

The planar multilevel interconnect process is accomplished by fabricating the first-level metal within dielectric windows and maintaining the metal thickness close to the dielectric thickness, resulting in a smooth planar surface greatly facilitating the fabrication of complex multilayer interconnects. Figure 14a shows an actual cross section of the planar crossover structure resulting from these process approaches. Figure 14b shows a top view of a similar structure. The process used to fabricate the planar multilayer structure is described as follows: an $\sim 5000\text{-\AA}$ plasma-enhanced CVD silicon nitride (plasma nitride) layer is deposited on the ICs. Then VIA windows are reactively ion-etched (RIE) through the ni-

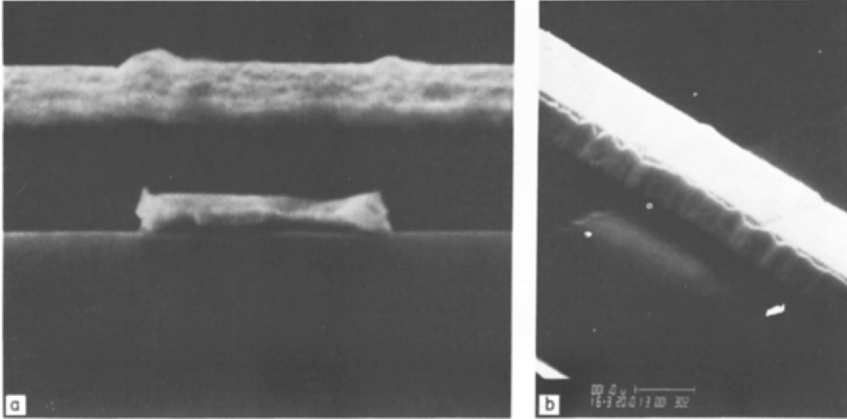


Fig. 14. (a) SEM cross section showing the planar multilevel interconnect structure illustrated in Fig. 13. (b) SEM photo showing the top view of a planar second-level interconnect overcrossing, isolated from the first-level interconnect by plasma nitride.

tride in order to interconnect the first-level metal to the second-level metal. A second-level metal composed of Ti/Au is deposited over the second-level dielectric and is subsequently defined by ion milling, completing the process.

Planar crossovers eliminate potential crossover problems such as shorts between first- and second-level interconnects and high resistances or open interconnects resulting from poor step coverage. Another very important advantage is that these planar structures allow both thinner metal and dielectric films to be used, resulting in less film stress, positively impacting yield and reliability. We also believe that these planar approaches have real potential for extension to three or more levels of interconnects, which will greatly impact the optimization of future complex VLSI.

More recently implemented into our process is the planar second-level VIA interconnect shown in the schematic of Fig. 13. A similar process to the enhanced lift-off method already described is used to fill the VIA windows with intermediate metal between the first- and second-level interconnects. Figure 15 shows SEM cross sections of a conventional VIA structure with its inherent step coverage limitations in contrast to the planar "Filled" VIA structure described here.

In this process, reactive ion etching is used to etch the VIA window with an anisotropic profile, followed by evaporated metal in order to fill the window. Any small voids or gaps (~ 500 Å) between the intermediate VIA metal and the dielectric are easily filled during the deposition of the

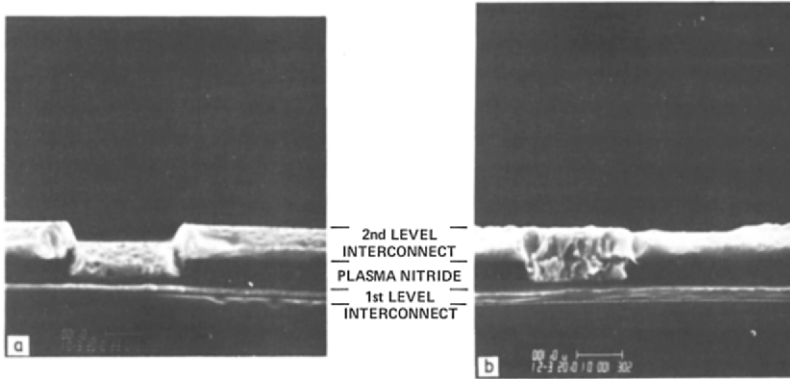


Fig. 15. SEM cross section comparing conventional VIA structures (a) to the planar second-level 'filled' VIA structure (b).

second-level metal. Figure 15b shows a SEM cross section of a filled VIA window along with the planar second-level interconnect. The motivation for developing this process is illustrated in Fig. 16, which contrasts two different VIA interconnect layouts, both having a common VIA hole cross-sectional area ($1.5 \times 1.5 \mu\text{m}$). Conventional processing and layout designs call for isotropic VIA side walls, which require larger geometries

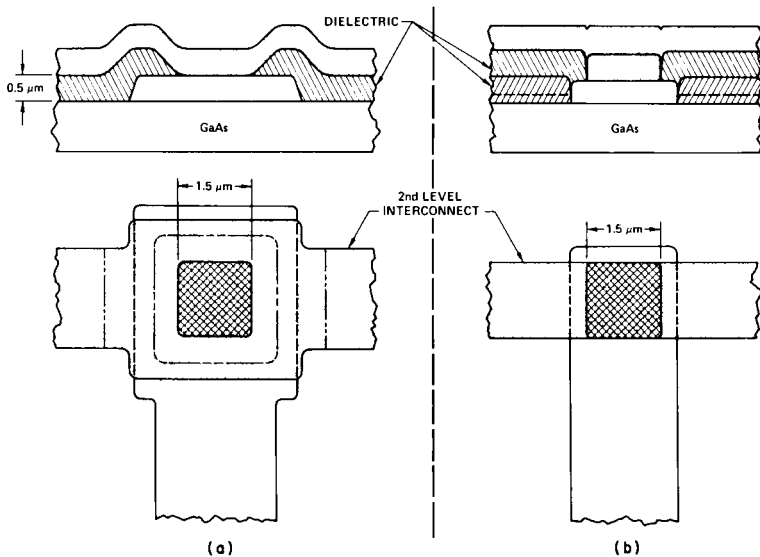


Fig. 16. Comparison of a conventional IC VIA layout versus a filled VIA layout design, illustrating the impact on density and alignment tolerance using these different approaches.

for the second-level interconnects and tight alignments for ensuring adequate step coverage. However, filled VIAs do not require any step coverage; therefore, smaller second-level interconnects with relaxed alignment tolerances may be utilized. The capability of achieving higher densities with increased process yields is evident. These VIA approaches are expected to favorably impact future VLSI requirements.

D. GaAs LSI/VLSI Process Status

The integrated circuit performance advantages that can be obtained by virtue of gallium arsenide's high electron mobility and aided by the availability of semi-insulating GaAs substrates have motivated the research in GaAs IC processing technology presented here. The results of this research have provided significant advances in GaAs processing techniques, allowing the full development of a planar, "silicon-like," LSI fabrication technology.

The planar-implantation microstructure pattern-replication techniques described here have been developed specifically for LSI/VLSI requirements. A key goal for this process development was to demonstrate the successful operation of a GaAs LSI circuit. This goal has been achieved through the successful fabrication and functional operation of the GaAs 8×8 -bit parallel multiplier containing over 1000 gates shown in Fig. 17. Testing of this circuit has verified full functionality for all 64,000 possible multiplier combinations, with NOR gate delays as low as 150 psec 65,536 measured. This gate delay corresponds to an ~ 5 nsec multiplication time to obtain a 16-bit product. These speeds are significantly higher than any currently available silicon ICs (see Section VI).

Process capabilities have been clearly demonstrated at the LSI level, as indicated by the component and feature count for this multiplier circuit, shown in Table V. The gate count of 1008 corresponds to over 6000 individual devices, $\sim 12,000$ crossovers, and ~ 7000 VIAs. This advanced planar GaAs 8×8 -multiplier LSI circuit has been made possible through the use of high-yield planar microstructure fabrication techniques. Demonstration of this LSI part provides us with excellent prospects for achieving future VLSI.

In summary, we have presented a GaAs fabrication process that is well on the way to meeting the goals of future GaAs VLSI. The excellent prospects of GaAs VLSI are enhanced by the fact the basic process steps used in fabricating GaAs ICs are generally fewer than those used in Si ICs. This is possible mainly because of the availability of semi-insulating sub-

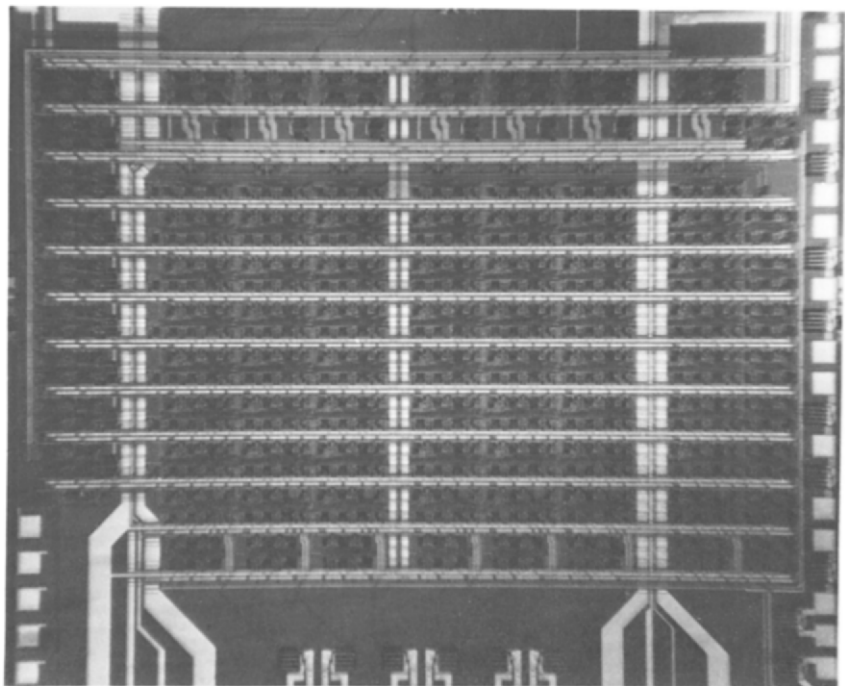


Fig. 17. Photograph of a planar GaAs LSI (1008-gate) 8×8 -bit parallel multiplier.

strates, providing minimal parasitic capacitances and direct device isolation without any additional complicated processes as found in Si NMOS, CMOS, or CMOS/SOS. These factors should have a significant favorable impact on the ultimate yield attainable for GaAs VLSI applications.

TABLE V

Component Count for a GaAs 8×8 -bit
Parallel Multiplier LSI Chip

COMPONENT	LOGIC CELLS (ADDERS)	PERIPHERAL (LATCHES, I/O)	TOTAL
LOGIC GATES	728	280	1,008
DIODES	2184	840	3,024
FETs	1456	560	2,016
ACTIVE LOADS	728	280	1,008
TOTAL DEVICES	4368	1680	6,048
OVERCROSSINGS	7400	4600	12,000
VIAS	4900	2100	7,000

VI. EXPERIMENTAL PERFORMANCE RESULTS FOR PLANAR SDFL LSI GaAs ICs

As will be indicated in Table VII, we have fabricated many different circuits, from SSI to LSI, to demonstrate the capability of this planar SDFL GaAs IC technology. The simplest demonstration circuits, very useful for establishing base-line gate propagation and dynamic switching energy values, are ring oscillators, which are formed of an odd number N_R of inverting gates (e.g., NAND, NOR, or inverter gates) [14]. Hence, with the proper logic state applied to inputs not in the ring signal path, no stable state exists for the ring, and the outputs must oscillate at a frequency determined by the gate delay τ_d [$f_{osc} = 1/(2N_R\tau_d)$]. Some of the ring oscillator results from our $L_g = 1 \mu\text{m}$ gate length planar SDFL gates are shown in Table VI. The highest speed obtained in low-power ($P_D = 1.1$

TABLE VI

Comparison of GaAs IC Ring-Oscillator Results for $L_g = 1 \mu\text{m}$ SDFL OR/NAND and SDFL NOR Gates

	GATE POWER	MODE	f_{osc} (MHz)	τ_d (ps)	$P_D\tau_d$ (fJ)
W = 10 μm OR/NAND GATE 7-STAGE RO FI = 4, FO = 2	$\left\{ \begin{array}{l} P_D = 1.7 \text{ mW} \\ V_{DD} = 2.34\text{V} @ 625 \mu\text{A} \\ V_{SS} = -0.86\text{V} @ -250 \mu\text{A} \end{array} \right\}$	$\begin{array}{l} G_1 \\ G_1 + G_2 \\ G_2 \end{array}$	$\begin{array}{l} 640 \\ 550 \\ 680 \end{array}$	$\begin{array}{l} 111.6 \\ 129.9 \\ 105.0 \end{array}$	$\begin{array}{l} 201.1 \\ 216.9 \\ 179.6 \end{array}$
W = 10 μm NOR GATE 9-STAGE RING OSC FI = 2, FO = 1	$\begin{array}{l} P_D = 865 \mu\text{W} \\ V_{DD} = 1.675\text{V} @ 460 \mu\text{A} \\ V_{SS} = -0.95\text{V} @ -100 \mu\text{A} \end{array}$		740	75	65
W = 5 μm OR/NAND GATE 7-STAGE RO FI = 4, FO = 2	$\left\{ \begin{array}{l} P_D = 475 \mu\text{W} \\ V_{DD} \approx 1.9\text{V} @ 200 \mu\text{A} \\ V_{SS} \approx -1\text{V} @ \sim -95 \mu\text{A} \end{array} \right\}$	$\begin{array}{l} G_1 \\ G_1 + G_2 \\ G_2 \end{array}$	$\begin{array}{l} 405 \\ 348 \\ 410 \end{array}$	$\begin{array}{l} 176.4 \\ 205.2 \\ 174.2 \end{array}$	$\begin{array}{l} 83.9 \\ 93.7 \\ 94.4 \end{array}$
W = 5 μm NOR GATE 9-STAGE RING OSC FI = 2, FO = 1	$\begin{array}{l} P_D = 372 \mu\text{W} \\ V_{DD} = 1.57\text{V} @ 218 \mu\text{A} \\ V_{SS} = -0.61\text{V} @ -48 \mu\text{A} \end{array}$		502	110.7	41.1
HIGHEST SPEED:					
W = 10 μm NOR GATE 9-STAGE FI = 2, FO = 1	$\begin{array}{l} P_D = 1.10 \text{ mW} \\ V_{DD} = 1.7\text{V} @ 546 \mu\text{A} \end{array}$		896	62	68
LOWEST $P_D\tau_d$:					
W = 3 μm INVERTER 9-STAGE FI = 1, FO = 1	$P_D = 110 \mu\text{W}$		817	136	16

^a The first four results listed were measured on the same chip of the same wafer for comparison.

mW) gates was $\tau_d = 62$ psec at $P_D\tau_d = 68$ fJ, while the lowest speed–power products were measured in very small geometry, $W = 3$ μm inverter gates with $P_D\tau_d = 16$ fJ at $\tau_d = 136$ psec. Of particular interest in Table VI is the comparison between the simple SDFL NOR gate (Fig. 4a) and the two-level SDFL OR/NAND gate (Fig. 4b) results for $L_g = 1$ μm ring oscillators of $W = 10$ μm and $W = 5$ μm sizes, respectively, measured on the same chip of the same IC wafer. These OR/NAND ring oscillators [17] were fabricated with FI = 4 gates (as Fig. 4b but with diode inputs B and G missing) with the output connected to both G_1 and G_2 inputs of the following stage (i.e., F connected to C and D in Fig. 4b). Hence, if a positive control line signal is applied to the common “A” inputs in Fig. 4b, oscillation through G_1 (the FET gate closest to the source) is obtained, while if the “A” inputs are low and the common “E” inputs are high, oscillation through the G_2 gates is obtained. If both the “A” input and “E” input control lines are low, then simultaneous propagation through G_1 and G_2 is required (the worst case because it simulates a 2- μm gate-length FET). As indicated in Table VI, the speed penalty for going to the two-level SDFL OR/NAND gates is quite modest, particularly for the $W = 10$ μm gate sizes. Hence, for circuit applications favoring the use of multilevel gate implementations, these configurations should be extremely attractive. For example, extrapolating the $\tau_d = 105$ –130 psec ring oscillator speeds to a fast complementary-clocked divider ($f_c \approx \frac{1}{2}\tau_d$) implemented with OR/NAND gates would give a maximum clocking frequency on the order of $f_c \approx 4$ GHz. As mentioned in Section IV, the use of three-level SDFL gates has been demonstrated for a $1\tau_d$ carry-delay ripple-adder circuits and would be attractive for the parallel multiplier as well. Of course, for many applications it would be desirable to eliminate the fan-in restrictions at the second and third levels of logic (typically FI = 2 for high circuit speeds) inherent in FET logic implementations, but with the unlimited diode fan-in available at the first logic level in SDFL, many very useful functions can be achieved.

While ring oscillators are useful devices for measurement of the basic speed–power properties of logic gates, a more realistic index of performance is the operation of gates in real sequential or combinatorial logic circuits. These circuits generally require the use of gates with multiple inputs and fanouts of two or more. For this study, circuits using D flip-flops were chosen as representative examples of sequential logic circuits; a parallel multiplier, implemented with half- and full-adders, was selected as a combinatorial logic circuit example. Data multiplexer and demultiplexer circuits with on-chip synchronous address generators, representing a mixture of sequential combinatorial functioning, were also fabricated [1,16,18].

TABLE VII

Review of Rockwell Planar GaAs IC Demonstration Circuits

FULLY WORKING, SUCCESSFUL TESTED CIRCUITS	
8 x 8 PARALLEL MULTIPLIER	1000 GATES
5 x 5 PARALLEL MULTIPLIER	260 GATES
8-STAGE SHIFT REGISTER/217 BIT P/N CODE GENERATOR	96 GATES
3 x 3 PARALLEL MULTIPLIER	75 GATES
4-BIT FAST RIPPLE CARRY ADDER USING 2- AND 3-LEVEL GATES	43 (63 EQUIV)
+ 10/11 OR + 20/22 VAR. MOD. DIVIDER FOR FREQ. SYNTHESIZER	37 GATES
8:1 DATA MULTIPLEXER WITH ADDRESS GENERATOR	64 GATES
1:8 DATA DEMULTIPLEXER WITH ADDRESS GENERATOR	60 GATES
1 GHz ANALOG SAMPLE/HOLD OR TRACK/HOLD CIRCUIT	25 COMPONENTS
DIGITAL FM DEMODULATOR CIRCUIT	65 COMPONENTS
+ 8 FREQUENCY PRESCALER (D-FLIP FLOP RIPPLE DIVIDER)	25 GATES
+ 8 SYNCHRONOUS COUNTER IMPLEMENTED WITH D-FFs	33 GATES
NOR GATE RING OSCILLATORS IN W = 3, 5, 10 AND 20 μm WIDTHS	9 GATES
OR/NAND GATE, 3 PROPAGATION MODE RING OSCILLATORS, 5 & 10 μm	7 GATES

Circuits in the MSI to 1000⁺-gate LSI range of complexity have been designed and fabricated using the planar SDFL approach (Table VII). All high-speed measurements required for testing the circuits have been made at wafer-probe level, although many packaged devices have also been tested, frequently with slightly better performance results. Output buffers

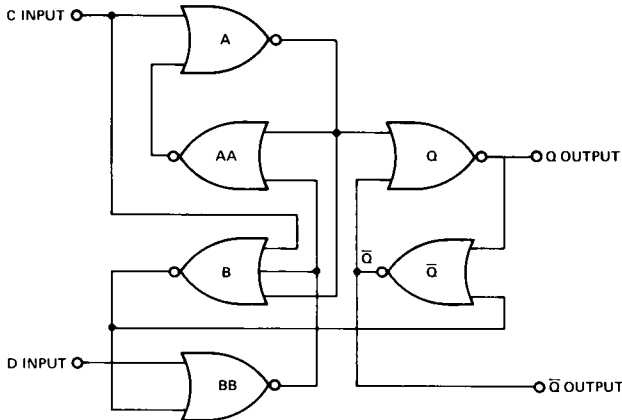


Fig. 18. Logic diagram for the NOR-implemented type D flip-flop used in various shift register and divider circuits in this work. In the ripple-divider applications, the D-FF is T-connected as a $\div 2$ by connecting the $\bar{Q}$ output to the D (data) input. In shift-register operation, the Q output is connected to the D input of the following stage.

of various types (source-follower, open-drain, etc.) are located on-chip to drive the output package capacitances or transmission-line interconnections, as required, to prevent loading of the internal circuitry. Binary ripple counters or frequency dividers have been made by using the D flip-flop (D-FF) implemented with six NOR gates, as a basic building block, as shown in Fig. 18. By connecting the D (data) input to the $\bar{Q}$ output, the clock input will produce an output transition for every full clock cycle ($\div 2$; this is called a *toggle*, or T-connected, D-FF). Logic simulation of this d-FF configuration indicates that correct operation should be obtained up to a clock frequency of $1/4.85\tau_d$, where τ_d is the propagation delay for the logic gates in the circuit.

A simple multistage binary frequency-divider configuration is the ripple divider, in which the Q output of the first T-connected D-FF is connected to the clock input of the second, etc. We have fabricated three-stage ripple dividers ($\div 8$) containing 25 SDFL NOR gates and have operated these up to a clock frequency of 1.9 GHz. Figure 19a shows “low-frequency” ($f_c = 100$ MHz) $\div 2$, $\div 4$ and $\div 8$ outputs, while Fig. 19b shows the $\div 8$ output at 237 MHz from this divider when the clock input is at $f_c = 1.9$ GHz. This corresponds to an equivalent propagation delay of 110 psec, which is in good agreement with ring oscillator data for 10- μ m SDFL NOR gates from the same wafer. The observed dynamic switching energy varied over the 0.25–0.45-pJ/gate range, depending on the bias conditions and wafer pinch-off voltage. Power dissipations were as low as 45 mW for the three-stage dividers [1,2,16].

In addition, larger MSI circuits, including an 8-input data multiplexer containing 64 gates, have been evaluated (logic diagram shown in Fig. 3 of Long *et al.* [16]). These circuits would be useful, for example, for parallel-to-serial conversion or a high-speed data transmission link. A one-input to eight-output data demultiplexer containing 60 gates was also fabricated and evaluated. Both circuits utilize a presettable three-stage synchronous counter [16] to generate the address for the multiplexer or demultiplexer gate arrays. (A ripple counter could not be used here since the outputs from the three stages are not “valid” at the same time; in the synchronous counter, where all the D-FFs are clocked simultaneously, all the Q and $\bar{Q}$ outputs form a valid address for the multiplexer.) A SEM photograph of a multiplexer chip is shown in Fig. 10.4 of Eden *et al.* [1] or Fig. 4 of Long *et al.* [16]; the chip size, excluding probe pads, is only 0.77×0.54 mm. Operation of both the multiplexer and demultiplexer chips has been demonstrated at a clock frequency of 1.1 GHz. Power dissipation of the multiplexer circuits varied from 75 to 375 mW for wafers with pinch-off voltages of $V_p = -0.5$ and -1.45 V, respectively.

In addition to the binary ripple and synchronous counters, we have fab-

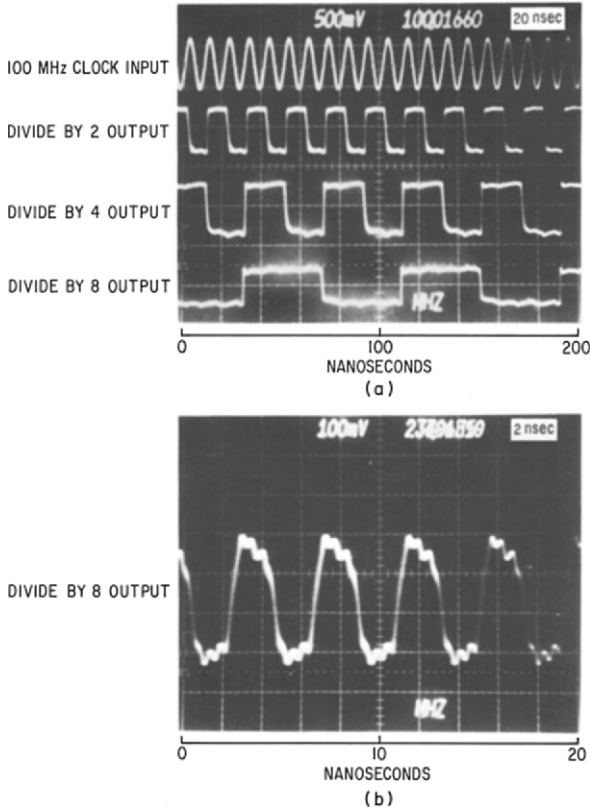


Fig. 19. SDFL D-FF binary ripple-divider results [1,16,17]: (a) Low-frequency operation showing 100-MHz sine wave clock and $\div 2$, $\div 4$ and $\div 8$ circuit outputs; (b) $\div 8$ output of this ripple divider operating at a clock frequency of 1.9 GHz.

ricated and tested at clock rates up to 1.4 GHz variable-modulus dividers ($\div 5/6$; $\div 10/11$, $\div 40/41$, $\div 10/12$, $\div 20/22$, and $\div 80/82$ ratios) of the type used in frequency synthesizers. We have also fabricated larger sequential logic circuits of the type used for code generation in, for example, spread-spectrum communications. As noted in Table VII, we have fabricated and successfully tested a 96-gate 8-bit shift register/217 bit P/N code generator, implemented with eight D-FFs of the type shown in Fig. 18. In the shift register, the Q output of a stage is connected to the D input of the succeeding stage and all flip-flops are clocked together. This chip is set up for various modes of shift-register operation (serial load, parallel preset, recirculate, etc.). In addition, an exclusive-or gate with inputs taken from the fifth and eighth stages and with its output connected to the serial input at the first stage, make the circuit into a pseudonoise (P/N) code generator. Depending on the initial state set in with the parallel

preset inputs, P/N codes of 217-bit, 31-bit, and 7-bit lengths may be obtained. Proper operation in the shorter code lengths can be verified by directly examining the P/N code generator output on an oscilloscope, but this is rather marginally practical for the 217-bit case. It has proven easier for these high-speed P/N code generator circuits to verify operation by examining the output on a spectrum analyzer (a Tektronix 7L13, 0–1860 MHz was used) for the proper overall amplitude envelope (as calculated from the Fourier transform of the P/N code) and the proper spacing of the frequency lines in the output spectrum (e.g., $F_c/217$, $f_c/31$, or $f_c/7$).

A particularly attractive candidate for a representative combinatorial logic part is a parallel multiplier, since multiplication frequently represents a bottleneck in signal-processing and computer systems. Typical sizes for a high-speed parallel multiplier would be 8×8 to 16×16 bits, with larger products formed of combinations of these. A straight parallel multiplier, $N \times N$ bits, without carry lookahead and not using more complex (e.g., Wallace tree) approaches, requires $N(N - 2)$ full-adders and N half-adders in its implementation and requires a total of $N - 1$ sum delays plus $N - 1$ carry delays to obtain the product. A goal of our ARPA GaAs LSI program was to construct an ~ 1000 -gate, 8×8 -bit parallel multiplier chip (760-gate multiplier plus I/O latching and buffering). As a first step toward this goal, a 75-gate, 3×3 -bit, parallel multiplier was fabricated, followed by a 260-gate, 5×5 -bit version. A fully latched 1008-gate, 8×8 -bit parallel multiplier was then designed and fabricated, with the first fully working devices achieved in late 1980.

Since the bulk of a large multiplier is full-adder cells, the speed of these cells dictates the multiply time. Figure 20 shows the logic diagram for a

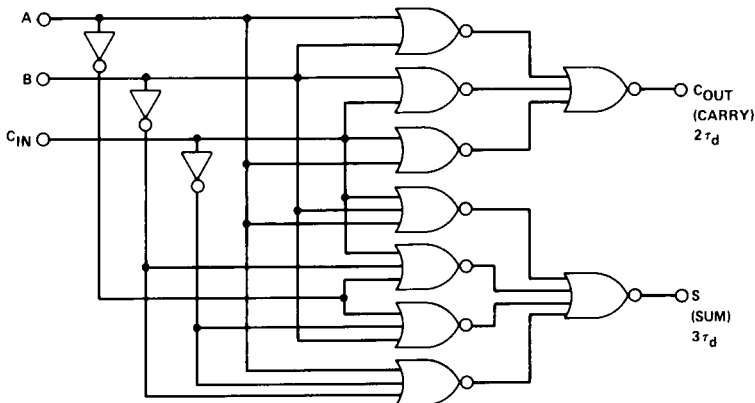


Fig. 20. Full-adder cell used for array multipliers, implemented with 12 SDFL NOR gates. Carry delay is $2\tau_d$ and sum delay is $3\tau_d$, where τ_d is the basic NOR gate propagation delay.

NOR-implemented full-adder cell. By using a full minterm expansion for both sum and carry, the propagation delays are kept to $2\tau_d$ for the carry and $3\tau_d$ for the sum (where τ_d is the NOR gate propagation delay). A total of twelve gates (nine NORs and three inverters) are required per full-adder cell. Figure 21 shows the logic diagram for the 5×5 -bit parallel multiplier chip (FA stands for a full-adder cell and HA for a five-gate half-adder cell). For this $N = 5$ case, five half-adders and fifteen full-adders are used. A microscope photograph of this multiplier GaAs IC chip is shown in Fig. 20 of Eden and Welch [2]. These 260-gate, 5×5 -bit par-

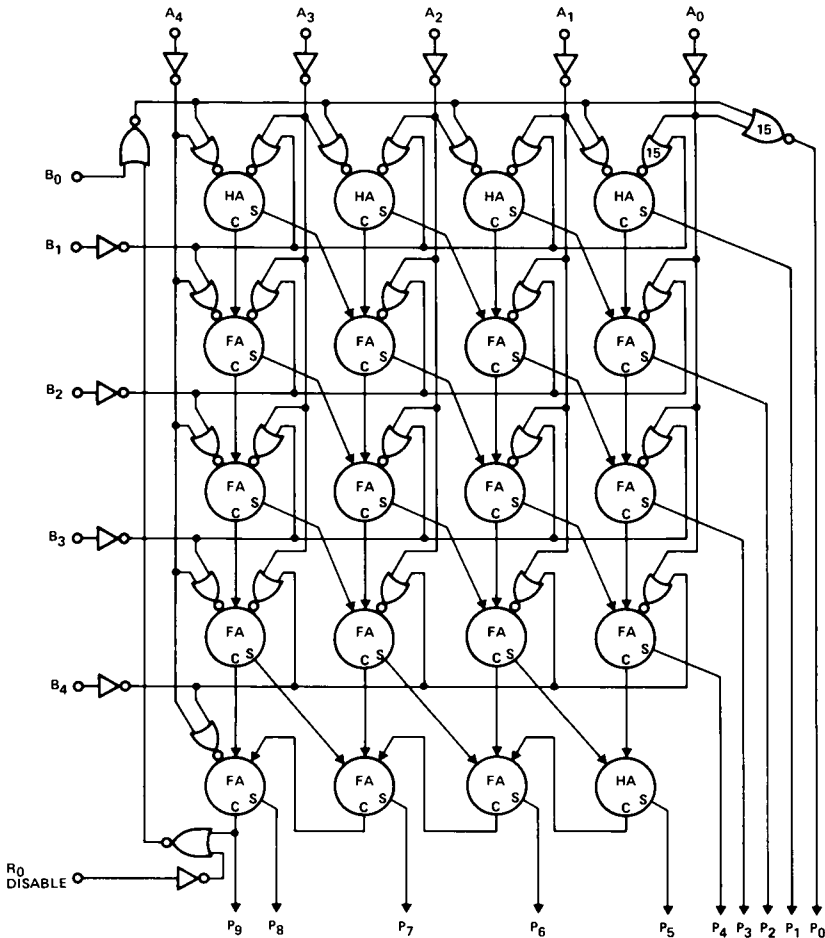


Fig. 21. Logic diagram of the 5×5 -bit parallel multiplier chip, where FA are full-adder and HA half-adder cells. Note the "ring oscillator" mode feedback path from P_9 to B_0 for testing propagation delay through the multiplier.

allel multiplier circuits have given good chip yields on GaAs substrate wafers grown by both the horizontal Bridgman and liquid-encapsulated Czochralski (LEC) methods (the LEC wafers had 3-in. diameters), with chip yields up to 20% obtained.

The 8×8 -bit latched parallel multiplier chip consists of a 760-gate parallel multiplier array plus input and output latches and drivers (248 gates). The 8×8 -bit parallel multiplier array itself is the same as shown for the 5×5 -bit multiplier in Fig. 21, except that it is expanded from 5 rows $\times$ 4 column to 8 rows $\times$ 7 columns. Unlike the earlier 3×3 and 5×5 multiplier chips, which consisted of the parallel multiplier array only, full input and output latching was incorporated on the 8×8 -bit multiplier chip. This makes it possible to use the multiplier in a pipelined mode with slower (silicon) systems, because a multiply can be in progress using the latched input operands while the outside system is disposing of the old latched 16-bit product and getting the new operands for the next multiply. The overall architecture for this GaAs SDFL 8×8 -bit parallel multiplier chip is shown in Fig. 22. Note that the input operand and/or output prod-

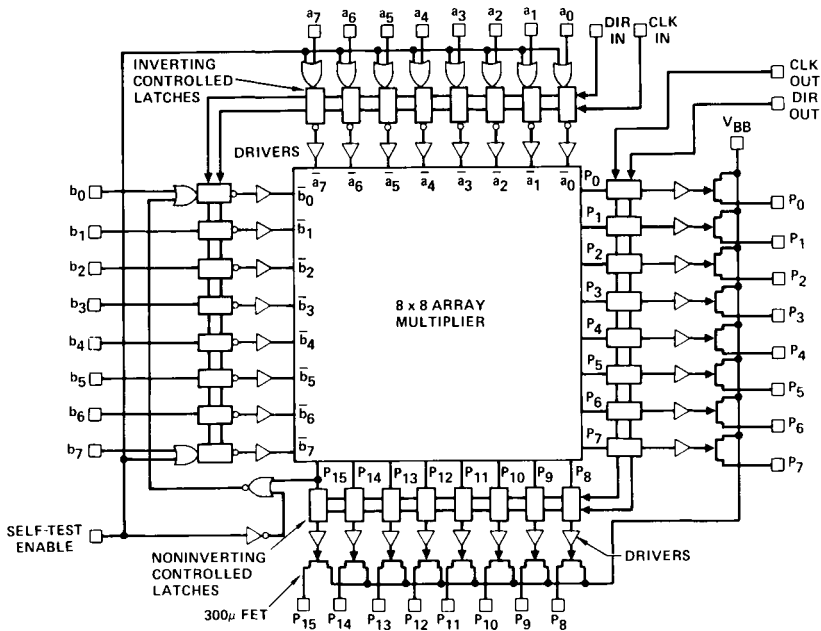


Fig. 22. Block diagram of the latched input/output 8×8 -bit parallel multiplier chip. The DIR IN and DIR OUT pins make the input and output latches transparent for direct parallel multiplier operation. The SELF TEST ENABLE pin enables the $\bar{P}_{15}$ to B_0 feedback path for "ring oscillator" mode MULTIPLY time measurements.

uct latches may be defeated for direct parallel multiplier array operation. Figure 17 shows a photomicrograph of the 1008-gate, 8×8 -bit multiplier chip. The orientation of the photograph is similar to the block diagram in Fig. 22, with input latches at the top and left, parallel multiplier array in the center, and output latches and drivers at the bottom and right. The overall chip size is 2.475×2.7 mm. While the bulk of the chip area is consumed in power supply busses on signal interconnection lines, even including these, the average gate density in the parallel multiplier array is 350 gates/mm² (~ 4 mil²/gate as compared to an actual gate area of ~ 1 mil²).

The wafer-probe test procedure for these multiplier chips allows for both static and dynamic performance measurements. Initial probe measurements for gross functionality involved the application of a specific input code with a square-wave signal on one of the inputs, which should produce a square-wave output, either in-phase or inverted, on all of the product bits (e.g., for the 3×3 , $111 \times 10S = \overline{SSSSS}$, where S is the square wave). If proper gross functionality is observed, complete low-speed functional testing is carried out for all possible combinations of operand inputs, using a MACSYM II computer system to generate inputs and measure and verify outputs (for all 65,536 combinations for the case of the 8×8 -bit multiplier). The problem of measuring MULTIPLY speed at wafer probe on such high-speed circuits was dealt with by adding a few extra gates to the multiplier chip to make an on-chip feedback path. For the 5×5 -bit multiplier (see Fig. 21), the invert of the most significant product bit P_9 may be applied back to the least significant multiplier bit B_0 . The input operands are selected as $A = 11111$ and $B = 1000B_0$, which gives the product $B_0\overline{B_0}B_0\overline{B_0}B_0\overline{B_0}B_0B_0\overline{B_0}B_0$; here the signal from the B_0 input to the $P_9 = B_0$ output (diagonally from upper left to lower right in Fig. 21 and then down and from right to left along the bottom row) represents the worst-case MULTIPLY time (3 sums + 5 carries) for the multiplier. Because after a MULTIPLY time is completed ($P_9 = B_0$), there can be no stable product when $\overline{P_9}$ is fed back to B_0 . Hence, in this feedback mode, the whole chip behaves like a very large ring oscillator, oscillating at essentially a frequency of $1/2T_M$, where T_M is the worst-case MULTIPLY time. (More precisely, $f_{osc} = 1/2N\tau_d$, where N is the total number of gate delays, including those in the feedback path; $N = 22$ for the 5×5 -bit circuit for $f_{osc} = 1/44\tau_d$, whereas $T_M = 21\tau_d$.) The feedback path for the 8×8 -bit latched parallel multiplier (Fig. 22) is similar, $\overline{P_{15}}$ to B_0 , but the feedback circuit goes through the input latch structure (which must be in its direct or "transparent" mode for proper operation). The SDFL NOR gate delays τ_d , measured on the 3×3 -bit multiplier chips, were $\tau_d = 172$ psec at $P_D = 750$ μ W/gate ($P_D\tau_d = 129$ fJ) or $\tau_d = 225$ psec at $P_D = 420$

$\mu\text{W/gate}$ for an excellent $P_D\tau_d = 95$ fJ dynamic switching energy, with similar results on the 5×5 s. The layout was improved somewhat on the 8×8 -bit multiplier chip, with gate delays as low as $\tau_d = 150$ psec measured, for a parallel MULTIPLY time ($T_M = 35\tau_d$) of $T_M = 5.25$ nsec at chip dissipation of about 2 W. At lower power levels (~ 1 W for the chip), the gate delays are typically $\tau_d \approx 200$ psec ($T_M \approx 7$ nsec).

The $T_M = 5.25$ nsec MULTIPLY time, corresponding to 190 million MULTIPLIES/sec, obtained with our GaAs SDFL 8×8 -bit parallel multiplier chips, is nearly an order of magnitude faster than commercial silicon bipolar parallel multipliers, at a fraction of the power dissipation. (A new ECL 8×8 -bit multiplier with $T_M = 19$ nsec at $P_D = 4.4$ W has been announced, but this chip uses a more exotic architecture and cannot, on an IC technology basis, be directly compared with these straight-forward parallel multiplier array results.) As discussed previously, the use of multilevel SDFL gates would reduce the full-adder propagation delay time from $3\tau_d$ sum, $2\tau_d$ carry to $2\tau_d$ sum τ_d carry, so that the 8×8 -bit parallel MULTIPLY time (6 sums and 8 carries plus τ_d) is reduced by 40% (from $35\tau_d$ to $21\tau_d$). This alone would increase, for the same $\tau_d = 150$ psec gate delay value, the MULTIPLY rate from 190 million multiplies/sec to 317 million MULTIPLIES/sec ($T_M = 3.15$ nsec). It is felt that, with modest further improvements in gate delay and improved multiplier architecture, 8×8 -bit MULTIPLY times of 2 nsec should be attainable (0.5 billion MULTIPLIES/sec).

VII. SUMMARY

In summary, the future prospects for using GaAs to achieve ultra-high-speed VLSI look very bright. Gallium arsenide logic circuits have shown ring oscillator gate delays as low as 30 psec at room temperature or 17.5 psec at 77 K using $L_g = 0.6 \mu\text{m}$ MESFETs. Other GaAs device approaches, such as the heterojunction bipolar transistor, might reduce these gate delays by an order of magnitude as the new processing technologies to fabricate these devices are mastered. As an indication of the present state of GaAs technology, we have developed in our ARPA/Rockwell GaAs LSI program, a super-planar-fabrication process capable of optimizing two or more types of devices as required for sophisticated LSI/VLSI circuit approaches. This process has demonstrated the capability for achieving tight parameter control on low-power DMESFETs and high-speed switching diodes and for achieving good yields on lower-complexity LSI (hundreds of gates) circuits (up to 20% on 260-gate, 5×5 -bit parallel multipliers). The Schottky diode-FET logic circuit ap-

proach used in this work has demonstrated high speeds (τ_d as low as 62 psec), low power ($P_D \approx 100 \mu\text{W}$ to 2 mW/gate), high densities (NOR gate areas as low as $600 \mu\text{m}^2$), and outstanding $P_D\tau_d$ products (as low as 16 fJ in ring oscillators or 95 fJ even in large MSI parts). The simplicity, high density, and large device parameter margins of SDFL, along with its speed and power characteristics, make it ideal for ultra-high-speed VLSI. We have demonstrated scores of different SDFL circuits, from SSI to LSI; some of these are listed in Table VII. We have fabricated 1008-gate LSI parts, 8×8 -bit latched parallel multipliers and have obtained fully working samples of these parts both on the smaller horizontal Bridgman-grown GaAs substrates and the new 3-in.-diameter LEC GaAs substrate material. We have demonstrated that even in these LSI parts the inherent GaAs speed is maintained, with SDFL NOR gate delays as low as 150 psec obtained in the 8×8 -bit parallel multiplier chips (corresponding to a 5.25-nsec MULTIPLY time).

Based on the results to date, extension of this planar GaAs IC technology to the capability for 10,000-gate VLSI circuits seems a reasonable goal, and efforts to achieve this goal are in progress.

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